# What does the Super Mario 64 courtyard plaque say?

A forward-model reconstruction of the "L is real 2401" texture: font, process and text

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**Abstract**

The stone plaque on the star statue in the castle courtyard of *Super Mario 64* (1996) has fuelled one of the longest-running legends in gaming: that it reads "L is real 2401". The plaque is a single 32×32 grey texture with two lines of text about three to five pixels tall, which is far below the resolution needed to read letters by eye. Instead of enhancing or interpreting the image, we *simulate* how a 1996 artist would have produced it. We type a candidate text in a real period font, resize it into the texture with a standard filter, and measure how well the result matches the stored pixels after removing the stone background. We do this for millions of candidate texts, fonts and process settings on a GPU, and calibrate the method on synthetic plaques with known text. We find that (i) the text is set in **Palatino Bold Italic**, most likely Adobe's own 1990 PostScript Type 1 version as used with Adobe Type Manager; it beats every other era font family under an identical, exhaustive process search, including the game's own dialog font, which is clearly ruled out; (ii) a *single rigid designer pipeline* reproduces the texture: 160×128 artwork, 23 px type, bilinear resize, slight blur and gamma 1.8. Two independent versions of Palatino converge on the same settings; (iii) line 1 reads **"Eternal Star"**: "Star" is the best of 27,122 words as the second word (whether the s is upper or lower case cannot be decided); (iv) line 2 is set in the same process at 17–18 px, centred under line 1, and "hope" is the best of 12,556 middle words. We then fitted line 2's own typesetting on the four letters of "hope" alone. On their own, held-out pixels, "We" becomes the best first word and "so" the best real ending, and the unconstrained dictionary search returns "We hope so" among its top three strings. With Adobe's original Palatino, a blind search over 17.7k words returns exactly "We hope so" as its best string. Our most likely reading is therefore

*Eternal Star*
*We hope so*

"L is real 2401", "A Secret star / lie here ~", "My hope is" and other published readings fit the pixels substantially worse under the same model. All code, results and figures are released for independent verification; no Nintendo data is redistributed.

## Contents

**1 Introduction** **3**

**2 The data** **3**

**3 Method: a forward model of the designer's process** **4**

**4 Identifying the font** **5**

**5 Reconstructing the process** **10**

**6 Line 1: "Eternal Star"** **12**

**7 Line 2: "We hope so"** **13**

**8 Working backwards from "hope"** **24**

**9 Calibration: how much can the pixels tell?** **30**

**10 The artist's font file: Adobe's 1990 PostScript Palatino** **32**

**11 Inverting the process: can the text be made visible?** **35**

**12 Comparison with published readings** **36**

**13 Limitations** **39**

**14 Conclusion** **39**

**A Reproducing the results** **39**

# 1 Introduction

In the courtyard behind Princess Peach's castle stands a statue topped by a star, with a small engraved plaque at its base (Fig. 1). The plaque is blurry, and for almost three decades players have argued about what it says.

- **"L is real 2401".** This is the classic reading from the late 1990s, taken as a hint that Luigi was hidden in the game [1, 2]. A variant spells the number as "2041" [3].
- **The 1998 Nintendo letter.** A Nintendo of America game-play counsellor's letter from 1998, which surfaced in 2016, called the plaque a developers' joke without saying what it reads [4, 5].
- **"Eternal Star".** This is the reading the Super Mario Wiki considers "more likely", set "in a heavily serifed font" [1].
- **"A Secret star / lie here ~".** Daniel Nodd (No64DD) proposed this in 2015 from photographs of CRT screens, and Nintendo Life covered it in 2019 [6].
- **"My hope is".** A further suggestion for line 2, based on two descenders and an ascender.

All previous attempts share one weakness. They *read* a 32×32 image, whether by eye, by upscaling or by photographing a CRT, and people then disagree about which shapes they see. At this resolution a human (or an upscaler) will find whatever text they are primed to find. This report takes the opposite approach.

1. **Model, don't read.** We write down how such a texture is made: type, resize, blur, gamma, on top of a stone background.
2. **Score hypotheses.** Every candidate (font, process, text) is rendered through that model and compared with the stored pixels.
3. **Calibrate.** Synthetic plaques with known text show what the method can and cannot recover.

We only use the texture as it ships in the game. Leaked source material is not used.

# 2 The data

The plaque is the texture `castle_grounds_textures.08800`. It is a 32×32 RGBA16 texture inside the MIO0-compressed `outside` texture segment, mapped onto a quad of about 142×105 world units in the courtyard model.

- We extracted it from the US, Japanese, European and Shindou ROMs, using segment offsets from the public decompilation [7].
- It is **byte-identical in all versions** (SHA-1 `88ef618a...`), so the text was never localised or revised.
- The texture is pure grey and uses only 23 distinct 5-bit levels.

There are two lines of text:

- **line 1:** rows 8–12, columns 2–29;
- **line 2:** mainly rows 17–19, columns 7–23;
- the lower rows are clean stone.

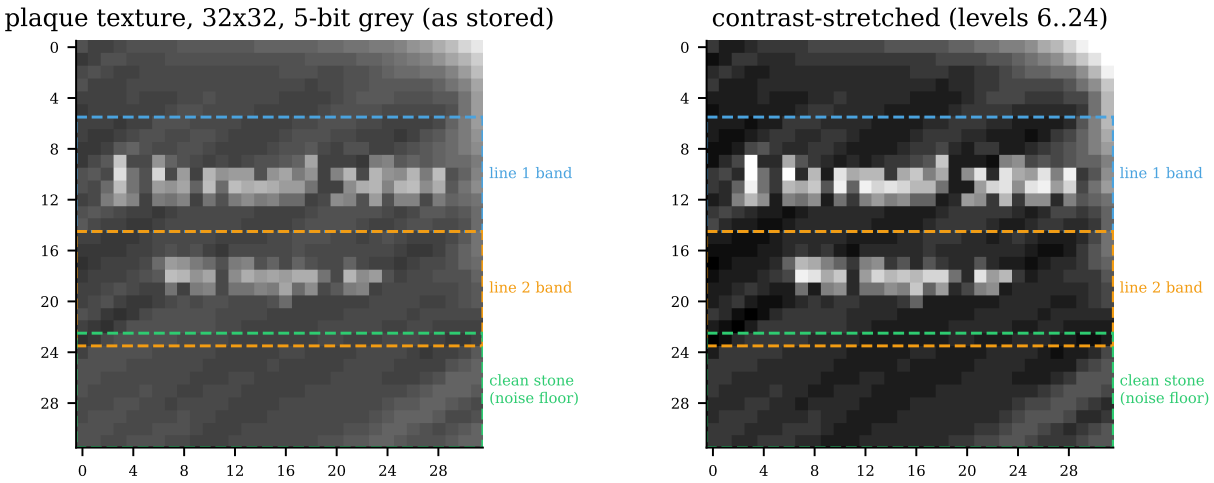



**Figure 1:** The plaque texture as stored in the ROM (left) and contrast-stretched (right). Dashed boxes mark the two text bands we model and the clean stone rows that define the noise floor.

## 3 Method: a forward model of the designer's process

**Why direct reading fails.** At 1.5–2 texture pixels per character, many different strings produce almost the same blur. In early experiments we let every letter move and change freely. Meaningless strings then fitted *as well as or better than* any real sentence, which shows that pixel evidence alone cannot identify the text. Identification only becomes possible once the *process* is constrained by the way the texture was actually made.

**Generative model.** We model the stored texture $t$ as

$$t \;=\; \text{stone} \;+\; L \cdot \big(\mathcal{R}(\text{text})\big)^{\gamma} \;+\; \text{noise},$$

where $\mathcal{R}$ is the designer's rendering process (Fig. 2):

1. The text is typed *once*, anti-aliased, with font $F$ at an integer pixel size $P$, with tracking, optional pair kerning, and fractional or whole-pixel advances.
2. It is typed on an artwork canvas of $32k_x \times 32k_y$ pixels.
3. The canvas is resized to $32 \times 32$ with a standard filter (nearest, box, bilinear, bicubic, Lanczos), whose support is scaled with the downsampling factor as in Photoshop and PIL.
4. An optional Gaussian blur is applied, then a gamma $\gamma$.

The engraving brightness $L$ is solved in closed form.

The stone is a nuisance. Inside each 4-column block of a text band it is modelled as a plane $(a + bx + cy)$ and projected out exactly, so the score is the residual sum of squares (SSE) after the best plane and the best ink level.

**Honest noise floor.** Applying the identical operator to the nine clean stone rows 23–31 gives SSE = 165.7. This is what pure stone texture with no text costs. No model can be expected to go far below a few times this value, because the stone under the letters is unknown.

**Search.** All renderings are expressed as matrix products: canvas→texture weight matrices, with every integer position as a slice. This makes exhaustive searches feasible on one consumer GPU (RTX 2060 S): millions of configurations per second for line 1, and a K-best dynamic programme over word sequences for line 2 (Section 7).




**Figure 2:** The designer pipeline we simulate. Every step is a single global setting. There are no per-letter adjustments.

# 4 Identifying the font

The first question is which typeface the artist used. We restricted ourselves to fonts that existed in 1995–96 and that a Nintendo artist could plausibly have had:

- Macintosh system and LaserWriter fonts: Apple System 7.5.3 Palatino and Times, New York;
- Adobe PostScript fonts: IRIX 5.3 Type 1, plus X11 Adobe bitmap fonts;
- Microsoft's 1996 core fonts;
- classic Mac bitmap fonts with QuickDraw styles;
- the game's own dialog font, extracted from the ROM;
- free metric-compatible clones of the 35 PostScript fonts (URW base35).

Modern fonts were excluded.

**Tournament (≈ 530 fonts and variants).** A flexible model first screened 393 font files, then refined the best 157. It was flexible in that it allowed scale, aspect, blur, gamma and small per-letter shifts. The best member of each family is shown in Fig. 3.

- **Palatino wins clearly** (554), ahead of Times (638), New Century Schoolbook (694) and Georgia (745).
- The SM64 dialog font (812), Helvetica/Arial (890) and all others are far behind.
- Within Palatino, the 1980s/1990s design (URW P052, Apple 1996, Adobe 1990) fits better than the revised Palatino Linotype of 1998/99, which scores 633.

**Identical rigid process search for every family.** The flexible model could favour fonts with odd spacing, so we repeated the comparison with the *rigid* designer model. Every font received exactly the same exhaustive search over the following settings:

- 4 canvas shapes and 10 sizes;
- tracking, kerning and two advance modes;
- 4 word spaces, 2 filters, 2 blurs and 4 gammas;
- all positions.

Figure 4 and Table 1 show the result. The best font is Palatino Bold Italic (Adobe Type 1, 1990) (SSE 733.3), and the best non-Palatino font, N. C. Schoolbook Bold Italic (URW), is 108 worse. Figure 5 shows what each family's best recipe actually prints: only the Palatino italics reproduce the narrow, slanted, high-contrast letters of the real plaque.

**The in-game font.** Many players assume the plaque uses *Super Mario 64*'s own dialog font. We tested this directly.

- **Setup.** The 71 glyphs and the game's letter widths are extracted from the ROM (US, segment 2, IA4). The line is typed in the game font and scaled up, either as pixel art (nearest neighbour) or with Photoshop's default bicubic. It then goes through exactly the same canvas, resize, blur, gamma and position search as every other font.
- **Result.** Its best "Eternal Star" error is 1288 on the comparison grid and 1279 with the full grid (7.8 times the noise floor), against 757–759 for Palatino Bold Italic. That ranks it 31 of 33 fonts.
- **Flexible tournament.** The game font also loses there (812 vs. 554).
- **Downstream decoding.** With the game font, a dictionary search for the second word does not return "Star" at the top: it was rank 86 in our earlier rounds.
- **Why.** The dialog font is a small bitmap font with one-pixel strokes, no serifs and no stroke contrast. The plaque's letters have the serifs and the thick–thin contrast of an old-style italic.
- **Verdict.** The plaque was *not* set in the game's font; it was typeset with a desktop font.

**Consistency checks.**

1. **Two independent Palatinos agree.** The free URW clone (P052) and the genuine Apple 1996 TrueType Palatino independently converge on the *same* process recipe (Section 5).

| # | font | family | SSE | Δ to best |
|---|---|---|---|---|
| 1 | Palatino Bold Italic (Adobe Type 1, 1990) | Palatino | 733.3 | +0.0 |
| 2 | Palatino Bold Italic (URW P052) | Palatino | 756.9 | +23.6 |
| 3 | Palatino Bold Italic (Apple 1996) | Palatino | 759.2 | +25.9 |
| 4 | Palatino Italic (URW P052) | Palatino | 801.6 | +68.3 |
| 5 | Palatino Italic (Adobe Type 1) | Palatino | 802.9 | +69.6 |
| 6 | Palatino Italic (Apple 1996) | Palatino | 809.1 | +75.9 |
| 7 | N. C. Schoolbook Bold Italic (URW) | New Century Schoolbook | 841.2 | +108.0 |
| 8 | Times Bold Italic (Apple 1996) | Times | 847.9 | +114.6 |
| 9 | N. C. Schoolbook Bold Italic (Adobe Type 1) | New Century Schoolbook | 882.6 | +149.3 |
| 10 | Times Bold Italic (URW Nimbus) | Times | 891.8 | +158.5 |
| 11 | Times Bold Italic (Adobe Type 1) | Times | 913.4 | +180.1 |
| 12 | Times Bold Italic (Adobe, IRIX 5.3) | Times | 913.4 | +180.1 |
| 13 | Georgia Bold Italic (MS 1996) | Georgia | 956.1 | +222.8 |
| 14 | Adobe Garamond Bold Italic (Type 1) | Garamond | 963.2 | +230.0 |
| 15 | Zapf Chancery (URW Z003) | Zapf Chancery | 971.6 | +238.4 |
| 16 | N. C. Schoolbook Italic (URW) | New Century Schoolbook | 984.5 | +251.3 |
| 17 | Times Italic (Apple 1996) | Times | 992.8 | +259.5 |
| 18 | Palatino Bold (URW P052) | Palatino | 997.0 | +263.7 |
| 19 | Zapf Chancery (Adobe Type 1) | Zapf Chancery | 1007.4 | +274.1 |
| 20 | Times New Roman Bold Italic (MS 1996) | Times | 1007.5 | +274.2 |
| 21 | Georgia Italic (MS 1996) | Georgia | 1030.4 | +297.1 |
| 22 | Palatino Bold (Apple 1996) | Palatino | 1033.8 | +300.6 |
| 23 | Palatino Roman (URW P052) | Palatino | 1056.3 | +323.1 |
| 24 | Palatino Roman (Apple 1996) | Palatino | 1074.3 | +341.1 |
| 25 | Bookman Light Italic (URW) | Bookman | 1136.5 | +403.2 |
| 26 | Bookman Demi Italic (Adobe Type 1) | Bookman | 1222.0 | +488.8 |
| 27 | Helvetica Bold Italic (URW Nimbus) | Helvetica / Arial | 1229.8 | +496.5 |
| 28 | Bookman Demi Italic (URW) | Bookman | 1232.4 | +499.1 |
| 29 | New York (Apple 1996) | New York | 1255.2 | +521.9 |
| 30 | Helvetica Bold Obl. (Adobe, IRIX 5.3) | Helvetica / Arial | 1263.6 | +530.3 |
| 31 | SM64 in-game dialog font (from ROM) | SM64 dialog font | 1288.0 | +554.7 |
| 32 | Avant Garde Demi Obl. (URW) | Avant Garde | 1486.7 | +753.4 |
| 33 | Courier Bold Italic (URW) | Courier | 2109.6 | +1376.3 |

**Table 1:** Font ranking under the identical rigid process search (SSE of the best recipe for “Eternal Star”).

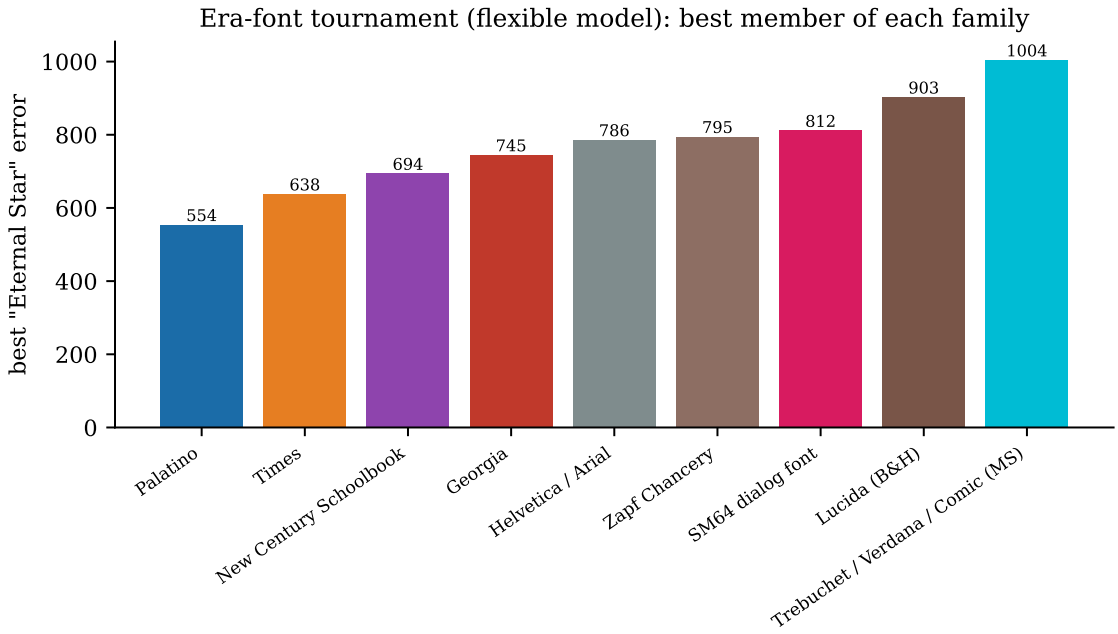



**Figure 3:** Era-font tournament, flexible model: the best "Eternal Star" error of each font family.

2. **Only Palatino decodes cleanly.** When the second word of line 1 or the words of line 2 are searched over a dictionary, Palatino returns "Star" and "hope" at rank 1. Times-family fonts return junk strings at the top, with "star" at rank 14–91.
3. **Weight and style.** Bold Italic and Italic are both close; upright Roman and Bold are clearly worse. The plaque is italic.

**Verdict.** The plaque is typeset in **Palatino Italic, most likely Bold Italic**, the classic Hermann Zapf design in its 1990s Mac/PostScript form. Among all Palatino digitisations, Adobe's original 1990 Type 1 file fits best; Section 10 identifies it as the artist's most likely font file.

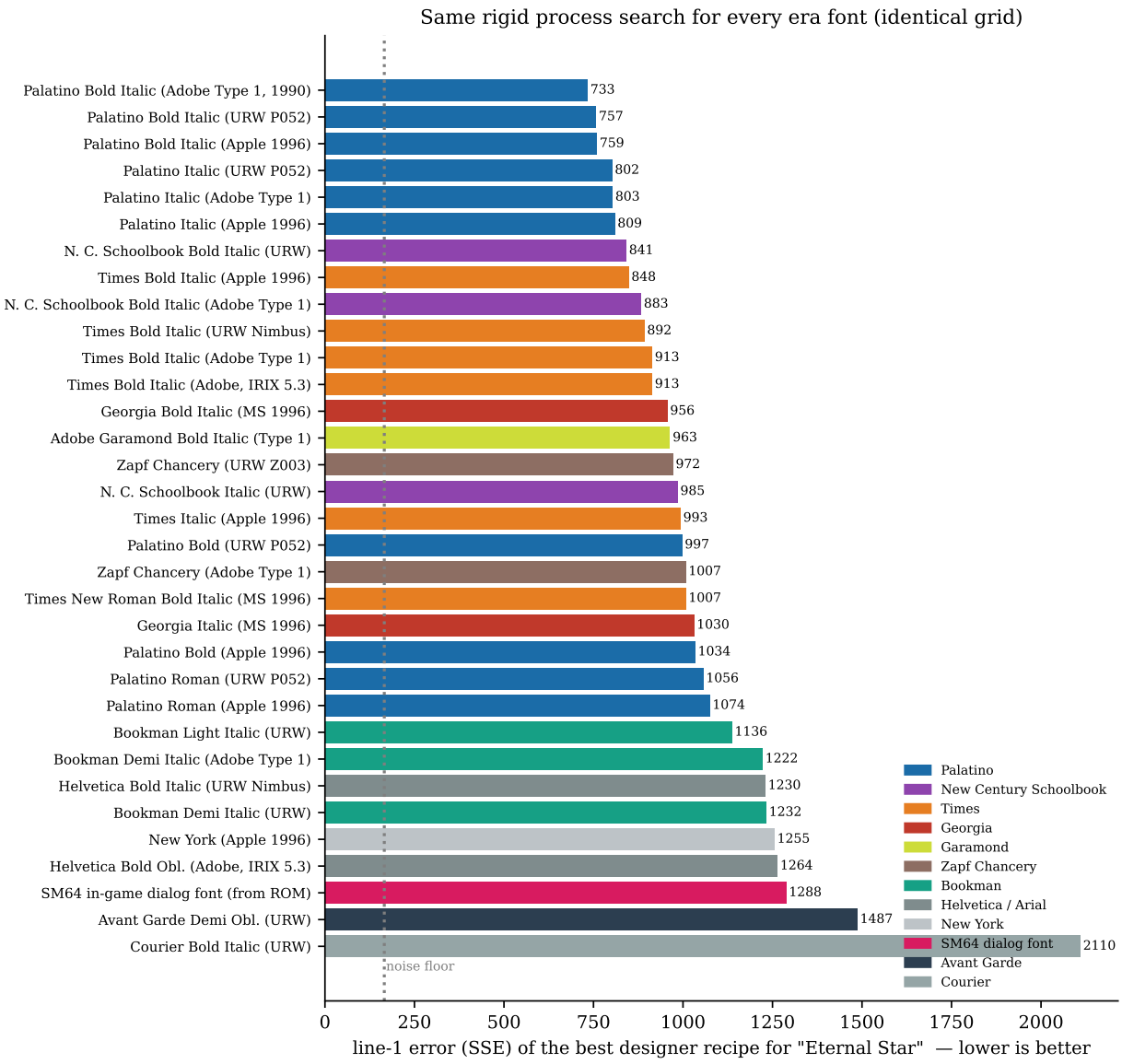



**Figure 4:** Same rigid process search for 33 era fonts: the best “Eternal Star” error per font, coloured by family.

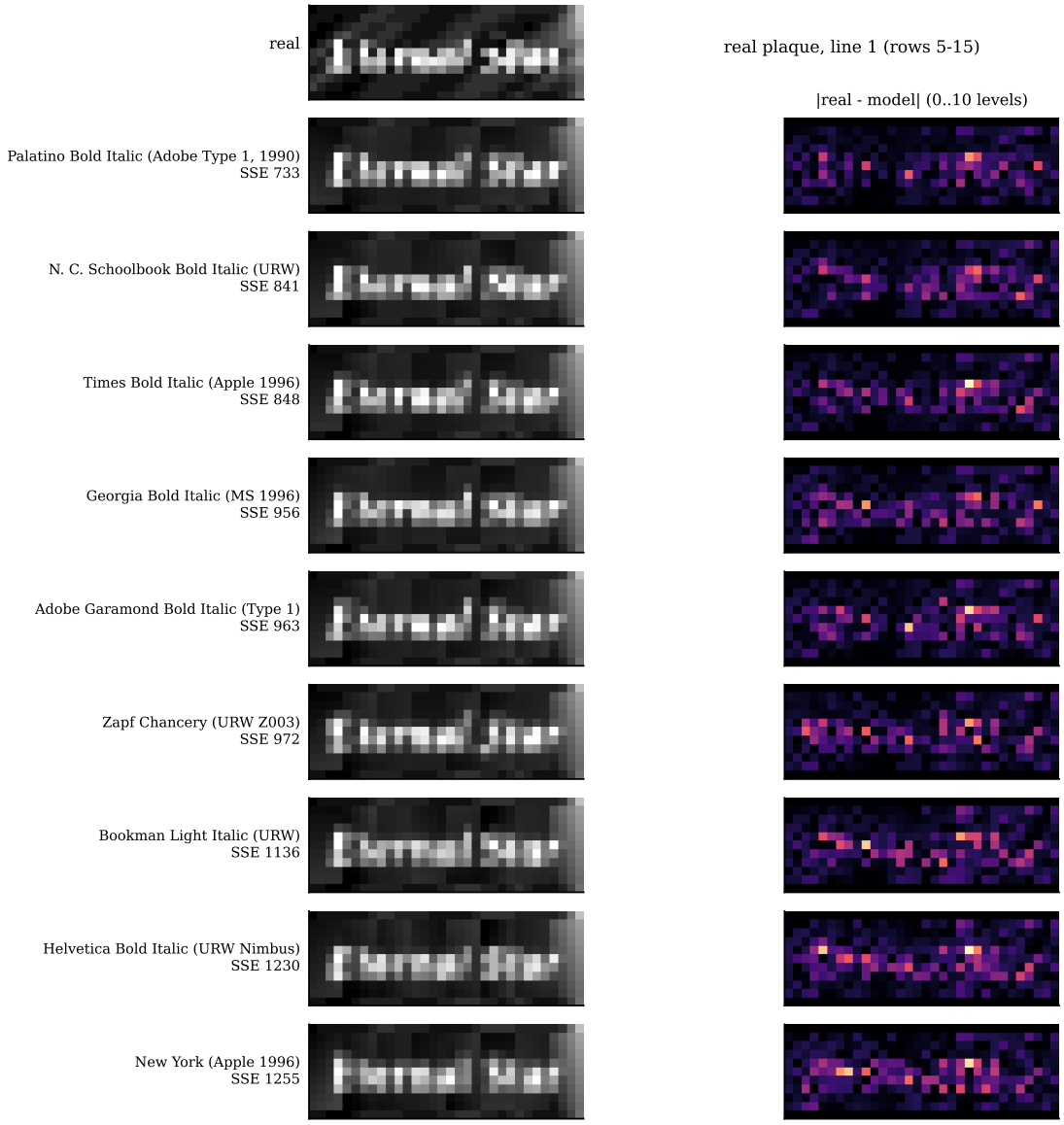



**Figure 5:** Line 1 as printed by the best recipe of each font family (left) and the absolute difference to the real plaque (right; 0–10 grey levels).

# 5 Reconstructing the process

Starting from a plain square canvas, we added one designer-plausible global step at a time (Fig. 6):

1. **Non-square artwork.** The art is drawn wider than tall and squeezed into the square texture. The quad itself is about 1.35:1.
2. **Word space, light blur and gamma.** The fit chooses $\gamma = 1.8$, the classic Macintosh display gamma.
3. **Whole-pixel advances.** Mac text without fractional widths places every letter on whole pixels.

The final recipe (Table 2) reaches SSE 757 (URW) and 759 (Apple), 4.6 times the noise floor. Both fonts choose the *same* canvas, size, tracking, filter, word space, blur and gamma.

**What is still missing?** A flexible model that may shift each letter horizontally reaches 548. With the per-letter shifts switched off, the same flexible model is *worse* than our rigid recipe (Fig. 7). The remaining gap is therefore pure letter spacing:

- "r n a l" sit about 0.3 px further right than these fonts place them;
- "Star" sits about 0.35 px to the left, which a narrower word space explains.

Eleven extra parameters on pure noise would buy about 6 SSE, not the 200 observed. The true 1996 spacing therefore genuinely differs slightly, through hinting by an unknown rasteriser or hand kerning. This affects neither the font family nor the words.

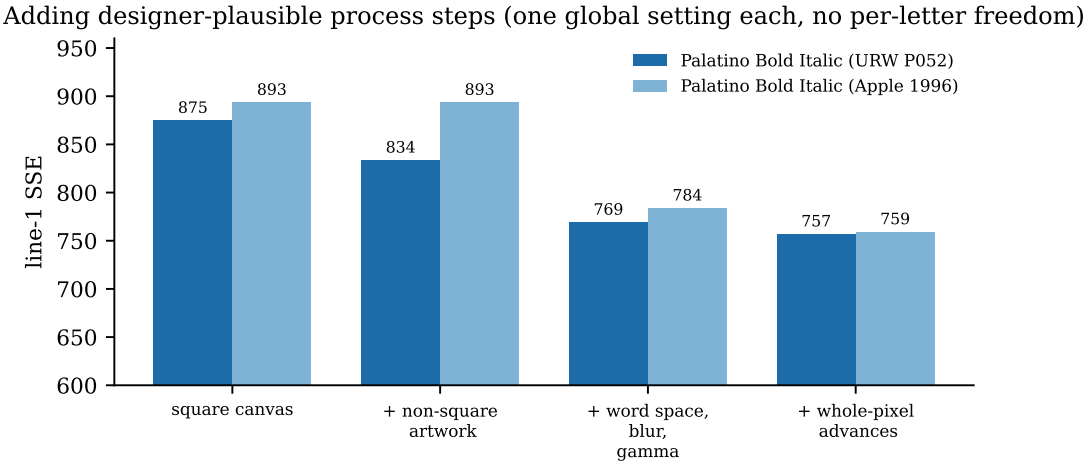



**Figure 6:** Each added process step is one global setting, chosen from a short list of options. The error drops from 875 to 757.

| | Palatino Bold Italic (URW P052) | Palatino Bold Italic (Apple 1996) |
|---|---|---|
| artwork canvas | 160×128 px | 160×128 px |
| font size | 23 px | 23 px |
| tracking | +1 px | +1 px |
| pair kerning | off | on |
| advances | whole pixels per glyph | whole pixels per glyph |
| word space | -0.5 texture px | -0.5 texture px |
| resize filter | bilinear | bilinear |
| blur after resize | $\sigma$ = 0.45 px | $\sigma$ = 0.45 px |
| gamma | 1.8 | 1.8 |
| line-1 SSE | 756.9 | 759.2 |
| SSE / noise floor | 4.57 | 4.58 |

**Table 2:** The reconstructed designer recipe for line 1.

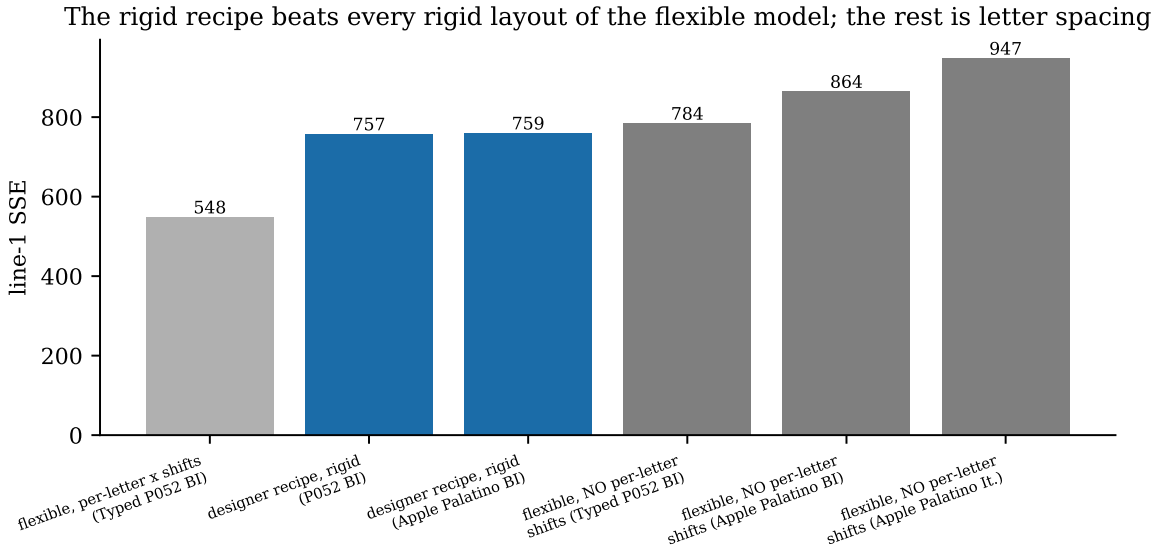



**Figure 7:** Ablation. The rigid recipe (blue) beats every rigid layout of the flexible model (grey); only free per-letter shifts do better.

# 6 Line 1: “Eternal Star”

**“Eternal”.** The first word is robust across every font, process and test. Its letters E, t and l are individually confirmed by local fits.

**The second word, with every setting frozen.** We froze the complete recipe (font, canvas, size, tracking, filter, blur, gamma and position) and rendered “Eternal ___” for each of 27,122 dictionary words. Only the word space (4 values) and the ink level remained free.

- “Star” is **rank 1 in both fonts** (Fig. 8).
- The runners-up (Stat, Sent, Seat) share its shape: S, a tall letter, two small letters.

**Fairness.** The recipe was tuned on “Star”, so we re-ran the *entire* recipe search separately for each rival (Fig. 9).

- “Star”/“star” still beat Stat, Sent, Slot and Seat.
- On line 1 alone, upper- and lower-case “s” are tied: 757 vs. 755 (URW) and 759 vs. 765 (Apple).
- **Joint two-line test.** One process must explain both lines. For each spelling we took its 100 best line-1 recipes and scored line 2 (“We hope so”) under each of them, then compared the best totals.
  - Genuine Apple font: “Eternal Star” 1132.5 vs. “Eternal star” 1176.3, so the capital S is ahead.
  - URW clone: 1101.9 vs. 1072.2, so the lower-case s is ahead.

  The two fonts disagree, so **the case of the s remains undecided**. We write “Eternal Star”, but the pixels do not settle it.

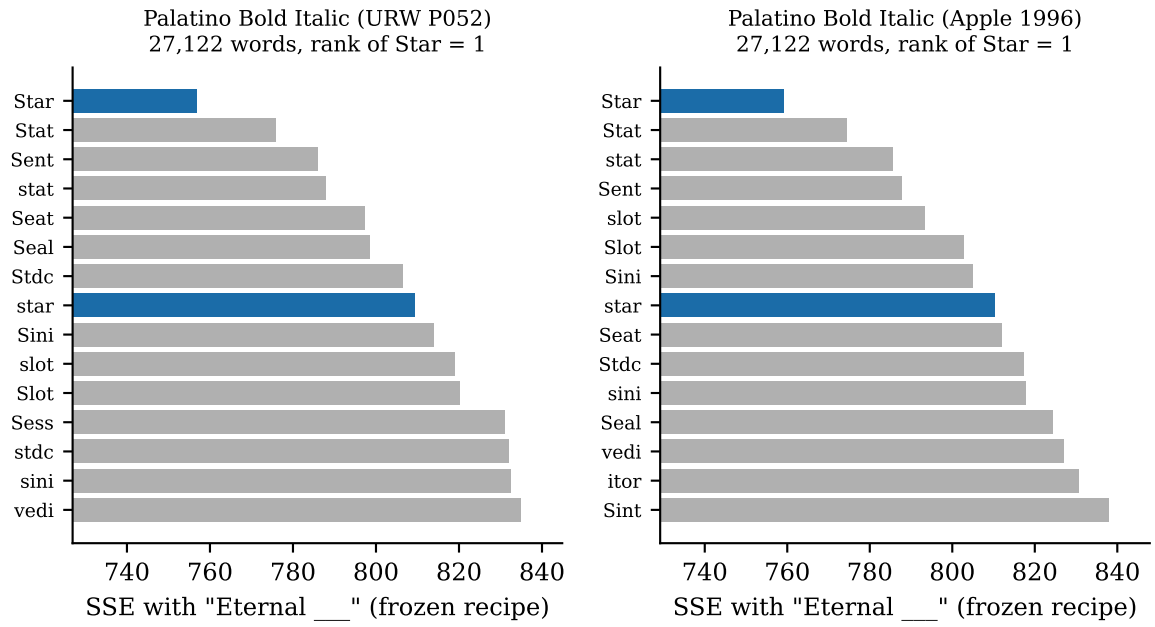



**Figure 8:** The second word of line 1 under the frozen recipe: top 15 of 27,122 words.

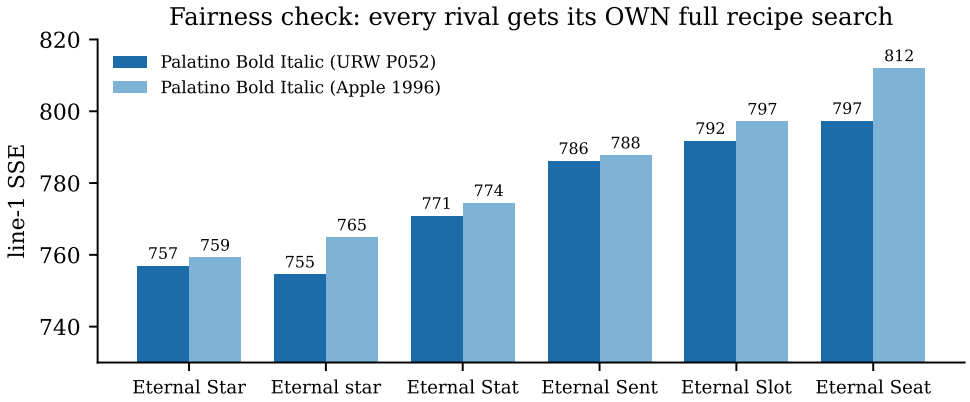



**Figure 9:** Every rival second word gets its own complete recipe search.

# 7 Line 2: "We hope so"

**Locked process.** We assume the designer set line 2 in the same document. It therefore shares the face, canvas, tracking, advance mode, filter, blur, gamma and *ink level* of line 1. This removes the contrast freedom that let nonsense win in earlier attempts. Only the size, the baseline, the start position and the word space are free.

**Fast exact-enough search.** Every glyph at every canvas position is pre-rendered through the process as a small "tile", and a string is a sum of tiles. A K-best dynamic programme then finds the best *word sequences*: any number of words, any start position, over a 17.7k-word English vocabulary. It runs for every size (13–20 px), baseline and word space. The top 3,000 phrases are re-rendered exactly.

**What comes out unprompted.**

- **Size and spacing:** 17 px, with exactly line 1's narrower word space.
- **Centring:** the line sits under line 1 (ink centre column 15.0 vs. 15.5), consistent with centred text.
- **Words:** 45 of the top 60 sequences contain the word "hope", and almost all begin with a W-word (Fig. 10).

**Exhaustive slot tests (Fig. 11).**

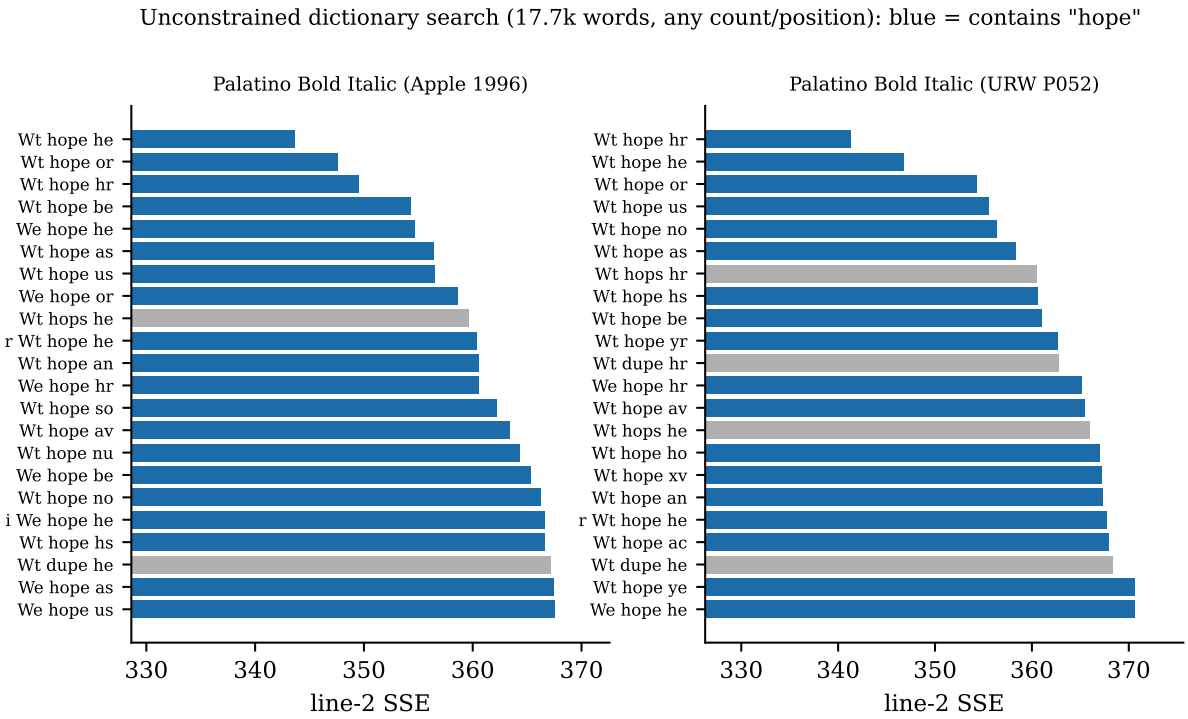



**Figure 10:** Unconstrained dictionary search for line 2. Without being told anything, "hope" appears in nearly every top sequence (blue).

- **Middle word.** In "We ___ so", "hope" is **rank 1 of 12,556 words in both fonts**, 16–19 SSE ahead of "hops"; everything else is more than 24 behind.
- **First word.** Among all 157k strings of one to three letters, the pixels want a capital W followed by one small letter. "We" is the best English word, ahead of "Me" and "He".
- **Ending.** The ending is two x-height letters. "he, or, us, as, be, no, an, so" lie within about 20 SSE of each other.

**The ending under line 1's typesetting.** Table 3 lists the main variants. The simulations in Section 9 show that two-letter endings within about 20 SSE cannot be separated by the pixels: a planted "We hope so" was decoded as "We hope an". Of the pixel-compatible endings, "so" is the only one that forms a complete sentence. Once line 2's own typesetting is calibrated (Section 8), "so" also becomes the best-fitting real ending.

Several alternatives are clearly worse:

- a Zapf Dingbats star, punctuation-only endings and Japanese-style "~" endings;
- "We hope you";
- Title Case "We Hope So";
- no ending at all, which scores 682.

**Descenders.** Row 20 lies just below the line-2 baseline. It is significantly brighter than the stone below it in *one* place only: column 16 (+5.0 levels; clean-stone s.d. 1.14), which is exactly where the "p" of "hope" falls (Fig. 13). A reading with two descenders, such as "My hope is", predicts a second descender under the "y" that the texture does not show.

## Close-up: candidates for "We" and for "so"

These are the two least certain words, so we isolated them. In each test the rest of the line is held fixed: either "___ hope so" or "We hope ___". Each candidate is rendered as a whole line with the locked recipe, but scored only on the stone blocks that contain the word in question:

- first word: texture columns 0–11;
- ending: texture columns 20–31.

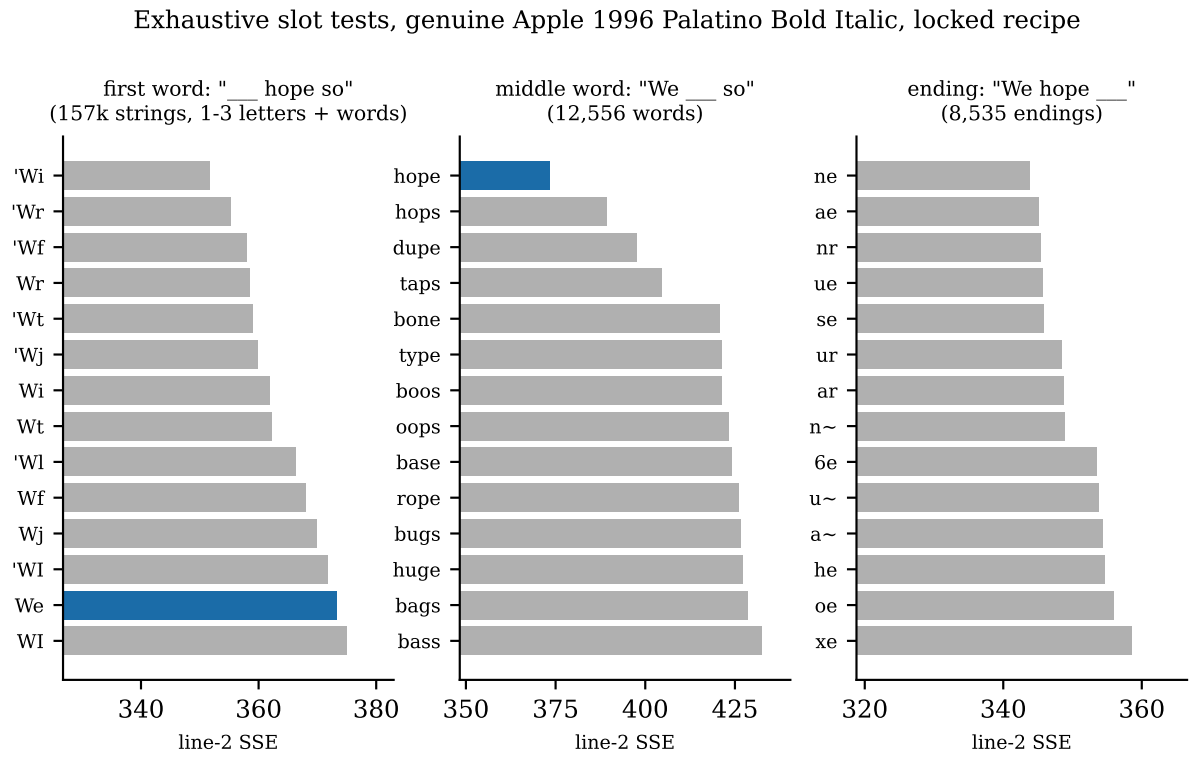



**Figure 11:** Slot tests with the genuine Apple 1996 Palatino Bold Italic: first word, middle word and ending.

Because each region is scored on its own pixels, the comparison is not diluted by the rest of the line. The best non-word strings found by the exhaustive search are shown for reference (Figs. 14 to 16).

**First word.**

- **What the pixels want.** A wide capital with a double-bowl or double-V top, followed by small letters. The non-words "Wi", "Wr" and "Wt" sit at the very top in both fonts.
- **Best real words.** "We'll", "We" and "By" are a tight group on their own region: "We" scores 187 (Apple) and 171 (URW), and "By" 176 (Apple).
- **Clearly worse in both fonts:** "Me", "He", "My", "No" and "Be".
- **Whole-line test.** A local fit is not enough, because the first word's width fixes where "hope so" must sit. On the whole line, "We'll" and "By" push "hope so" out of place and fail (Section 8); only "We" remains.

**Ending.**

- **Shape.** The ending is two x-height letters, with a bright stroke at the start and a rounded second letter. The best non-words ("ne", "nr", "ae") show exactly this.
- **Ranking of real endings.** In both fonts: "he" (113 / 114), then "or", "us", "as", "be", "no" (the order varies slightly between fonts), then "so" (132 / 134).
- **Clearly worse:** "to", "for" and "is". Their ascenders and the long "for" add ink where the texture has none.
- **What can be resolved.** The spread from "he" to "so" is about 20 local SSE. The simulations (Section 9) show that differences of this size between two-letter endings cannot be resolved: a planted "We hope so" came back as "We hope an".

**Variants.** Table 3 and Figs. 17 and 18 compare whole-line variants.

- The four best-fitting lines are "We hope he", "... or", "... us" and "... no". None of them is a complete sentence.
- **The best-fitting *complete* sentence is "We hope so"**, in both fonts. It is followed by "We hope so." and "We hope on".
- "We hope to", "We hope is", "We hope for", "My hope is", "He hopes so" and "I hope so" are clearly worse.

| line 2 | Apple | URW | mean | complete sentence? |
|---|---|---|---|---|
| “We hope he” | 355 | 371 | 363 | no |
| “We hope or” | 359 | 378 | 368 | no |
| “We hope us” | 368 | 379 | 373 | no |
| “We hope no” | 377 | 380 | 379 | no |
| “We hope so” | 373 | 389 | 381 | yes |
| “Me hope so” | 392 | 385 | 389 | no |
| “We hope so.” | 386 | 392 | 389 | yes |
| “We hope on” | 382 | 408 | 395 | yes |
| “We hope to” | 397 | 416 | 406 | no |
| “We hope so!” | 411 | 407 | 409 | yes |
| “We hope is” | 404 | 421 | 412 | no |
| “My hope so” | 435 | 410 | 423 | no |
| “He hope so” | 441 | 429 | 435 | no |
| “We hope for” | 460 | 434 | 447 | no |
| “No hope so” | 461 | 464 | 463 | yes |
| “My hope is” | 481 | 459 | 470 | yes |
| “We hope it” | 467 | 477 | 472 | no |
| “my hope is” | 489 | 475 | 482 | yes |
| “We hope all” | 510 | 461 | 485 | no |
| “We Hope So” | 523 | 481 | 502 | yes |
| “We hope one” | 514 | 492 | 503 | no |
| “We hope too” | 532 | 484 | 508 | no |
| “’Tis hope so” | 554 | 471 | 512 | no |
| “To hope is” | 523 | 538 | 530 | yes |
| “We hope for it” | 550 | 524 | 537 | yes |
| “We hope ~” | 522 | 585 | 553 | no |
| “We hope you” | 564 | 587 | 576 | no |
| “We hope to see” | 647 | 628 | 638 | yes |
| “We hope” | 682 | 696 | 689 | no |
| “We hope!” | 716 | 720 | 718 | no |
| “He hopes so” | 742 | 743 | 742 | yes |
| “I hope so” | 1047 | 1024 | 1035 | yes |

**Table 3:** Exact line-2 errors (whole band) of the main variants in both fonts, and whether each forms a complete sentence. Line-2 noise floor 165.7; stone only (no text) 2172.8.

**Other first words and other sentences.** We also tried alternatives suggested by the first-word close-up, and sentences that would make good sense on such a plaque, including quirky Japanese-English phrasings a 1996 Japanese team might have written.

- **Seeded search.** The global dictionary search was seeded with the first words We, By, Be, Me, My, He (and variants), with everything after them free. A further 91,884-phrase set combined these first words with about 90 middle words and 40 endings, in sentence and Title case.
- **“hope” survives every first word.** The best line for each first word is: “We hope he” (355), “Me hope he” (374), “He hope he” (389), “My hope he” (417), “By hope or” (420) and “Be hope us” (459), all in Apple Palatino.
- **“We” gives the best fit by a wide margin.** “By” matches its own region well, but its width pushes “hope” out of place.
- **Other sentences are far outside the pixel-compatible window** (Table 4, Fig. 19): “Be with us” and its variants, “By your side”, “We love you”, “Be happy”, “We pray so” and “We hope you”. None of them reproduces the tall-first-letter, mid-word-descender shape of the middle word.




**Figure 12:** Line 2 band: real pixels, model and difference for several readings.

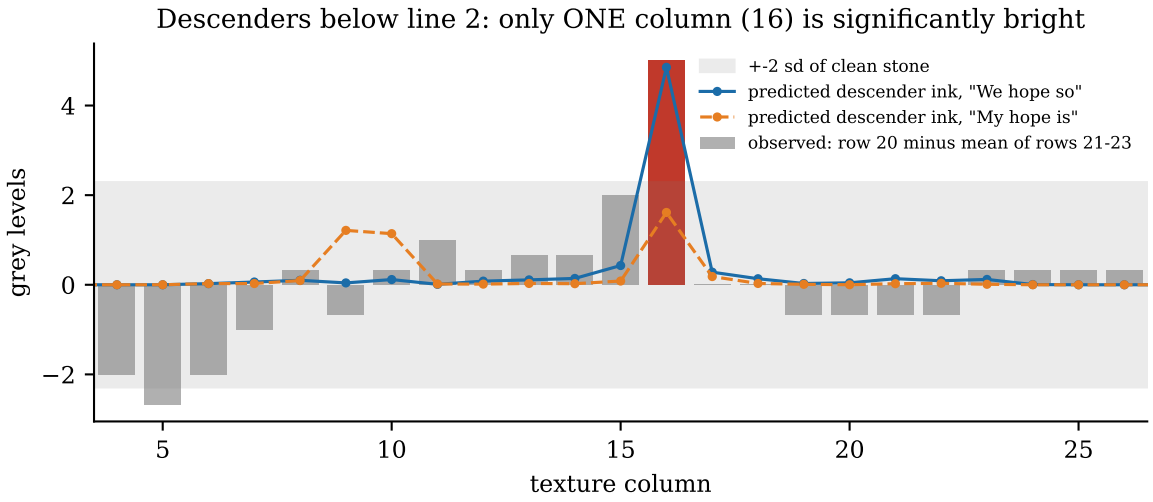



**Figure 13:** Descender evidence. Bars: brightness of row 20 relative to the stone below. Lines: descender ink predicted by "We hope so" and "My hope is".

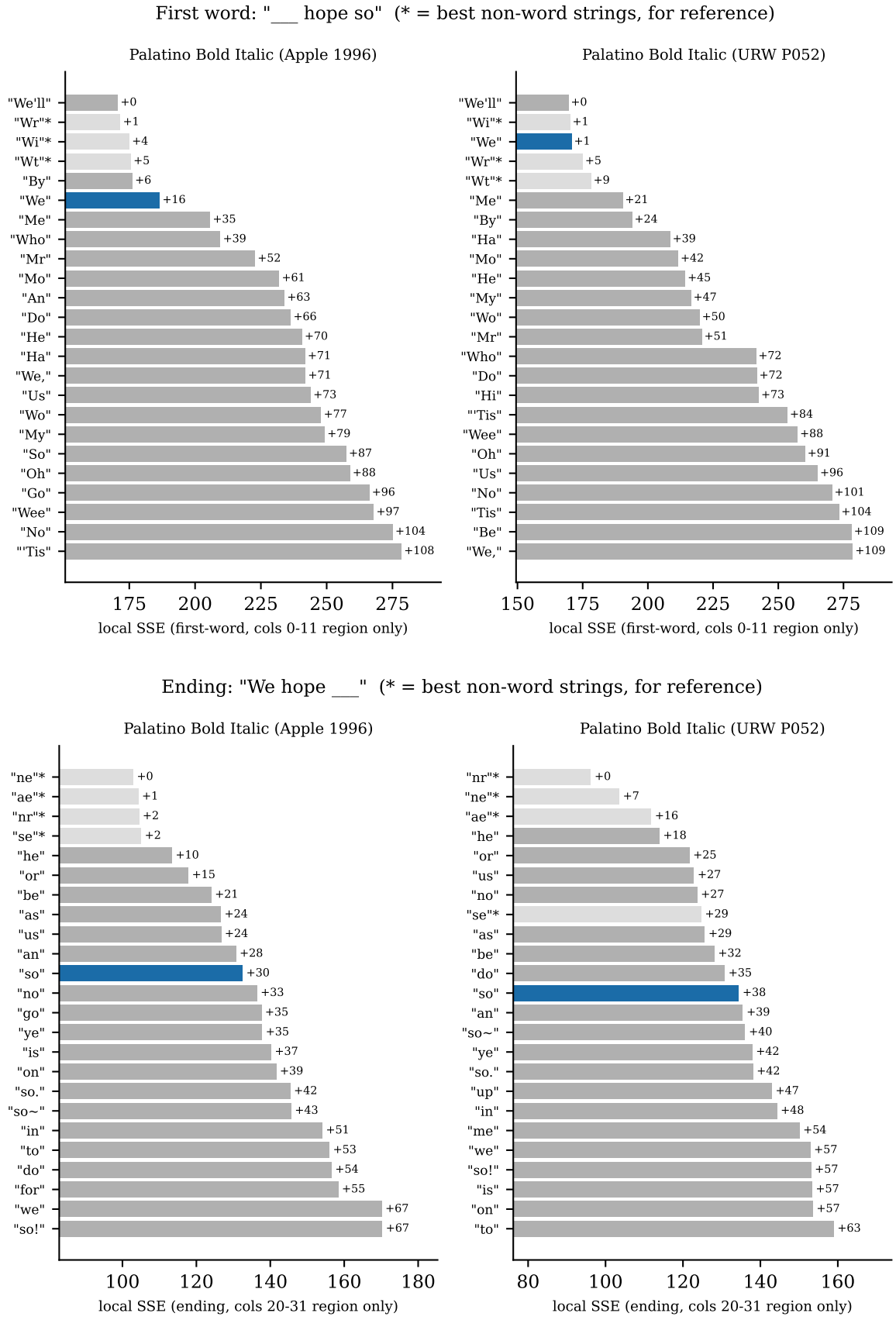



**Figure 14:** Local errors of first-word candidates (top) and ending candidates (bottom) in both fonts. Blue: our reading. Light grey with *: best non-word strings, for reference.

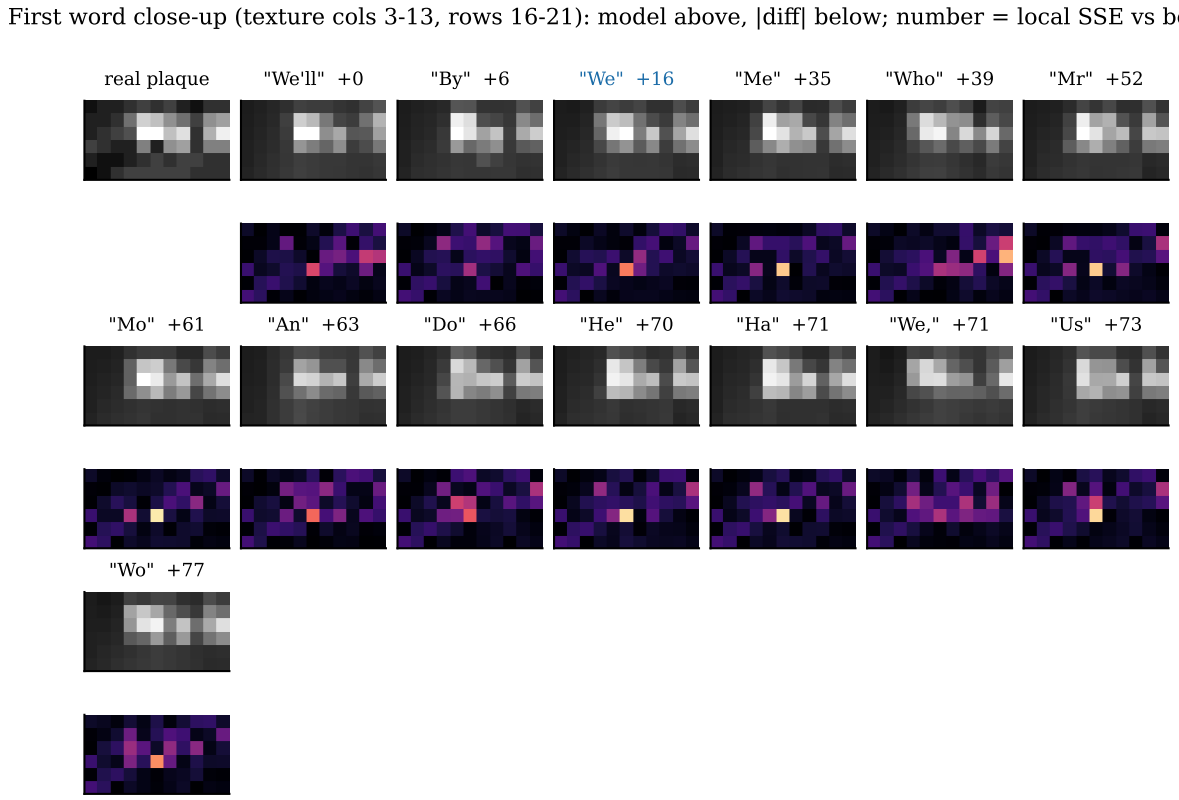



**Figure 15:** Close-up of the first word (Apple Palatino). For each candidate, the model of the zoomed region is shown above and the absolute difference to the real plaque below. The number is the local SSE relative to the best candidate.

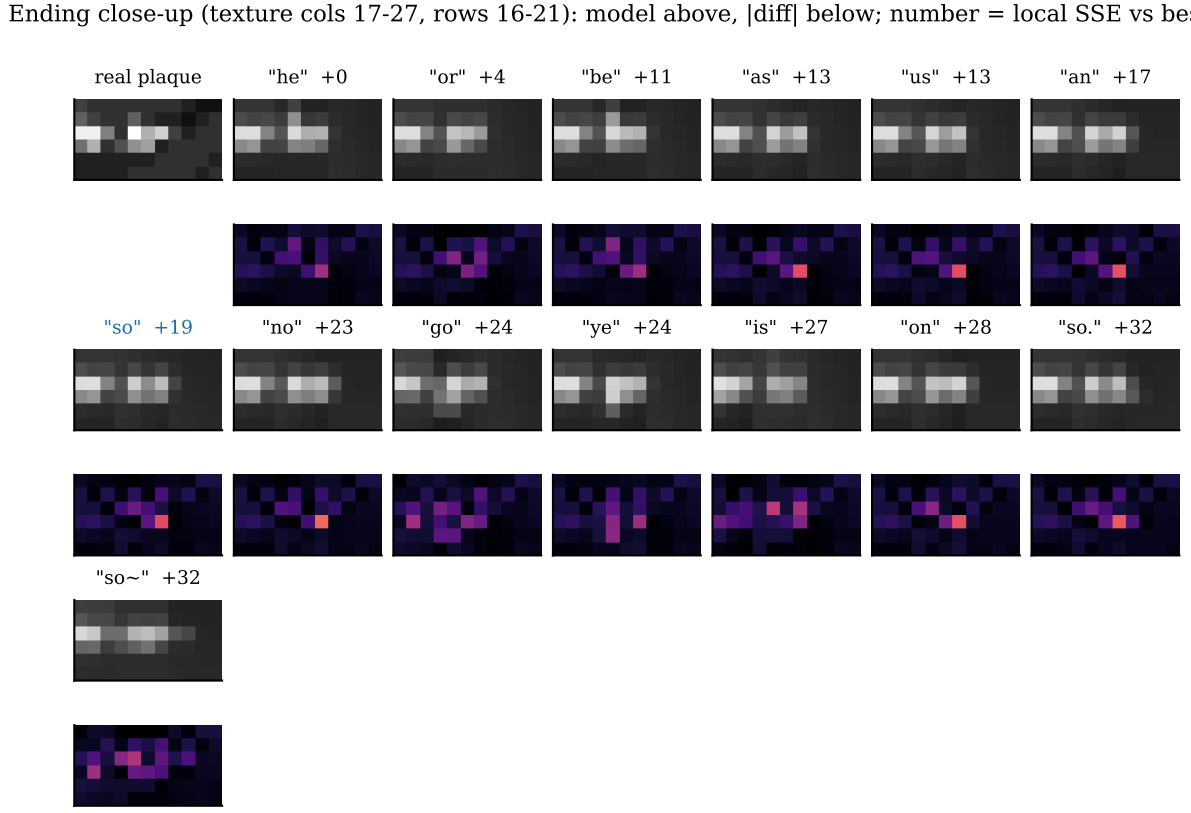



**Figure 16:** Close-up of the ending (Apple Palatino), laid out as in Fig. 15.

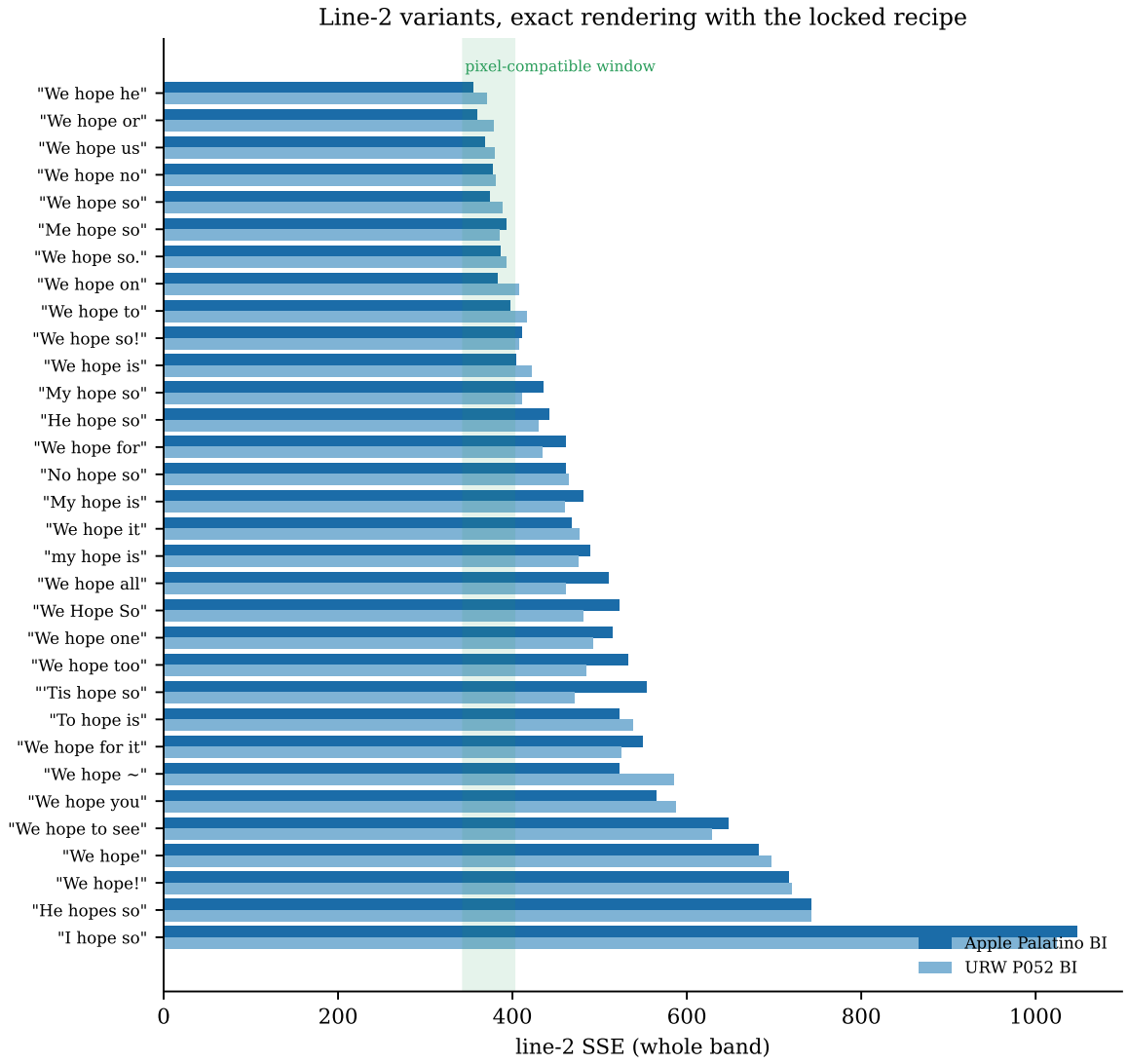



**Figure 17:** Whole-line variants in both fonts. The green band is the pixel-compatible window (Section 9).

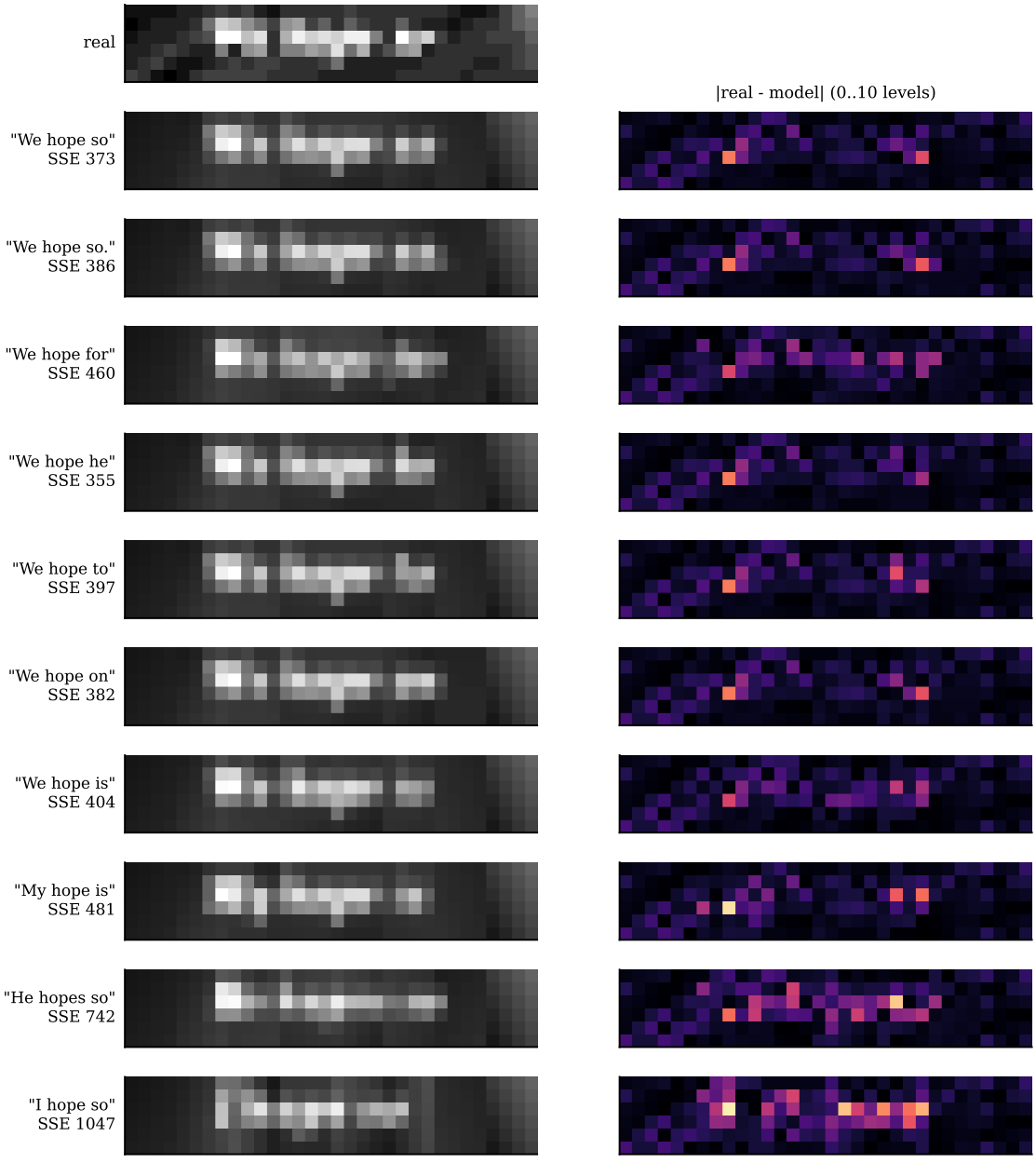



**Figure 18:** Line 2 for the main variants: model (left) and absolute difference to the real plaque (right).

| line 2 | Apple | URW | mean |
|---|---|---|---|
| “We hope he” | 355 | 371 | 363 |
| “We hope so” | 373 | 396 | 385 |
| “By hope so” | 434 | 409 | 421 |
| “Be hope so” | 492 | 493 | 493 |
| “We pray so” | 550 | 537 | 543 |
| “We hope you” | 552 | 587 | 569 |
| “We wish so” | 675 | 555 | 615 |
| “By my side” | 653 | 654 | 654 |
| “We hope it so” | 695 | 662 | 679 |
| “Be my star” | 681 | 685 | 683 |
| “Be with us” | 730 | 666 | 698 |
| “Be with me” | 724 | 675 | 699 |
| “Be with us~” | 731 | 669 | 700 |
| “Be happy so” | 705 | 726 | 716 |
| “Be with us!” | 752 | 689 | 720 |
| “Be happy” | 730 | 720 | 725 |
| “We love you” | 708 | 751 | 729 |
| “Be With Us” | 780 | 736 | 758 |
| “By your side” | 760 | 766 | 763 |
| “We are hope” | 752 | 793 | 772 |
| “Be my hope” | 794 | 771 | 783 |
| “We keep hope” | 764 | 811 | 788 |
| “Be here now” | 837 | 796 | 817 |
| “Be at peace” | 873 | 798 | 835 |
| “Be with you” | 867 | 858 | 863 |
| “Be our hope” | 886 | 861 | 873 |
| “Be here” | 889 | 865 | 877 |
| “Wish so” | 906 | 926 | 916 |
| “Be safe” | 927 | 916 | 921 |
| “We be ok” | 933 | 924 | 928 |
| “Hope so” | 1000 | 947 | 973 |
| “Be well” | 1062 | 1053 | 1057 |
| “Be ok” | 1297 | 1264 | 1280 |

**Table 4:** Alternative sentences for line 2 (exact, locked recipe; lower is better).

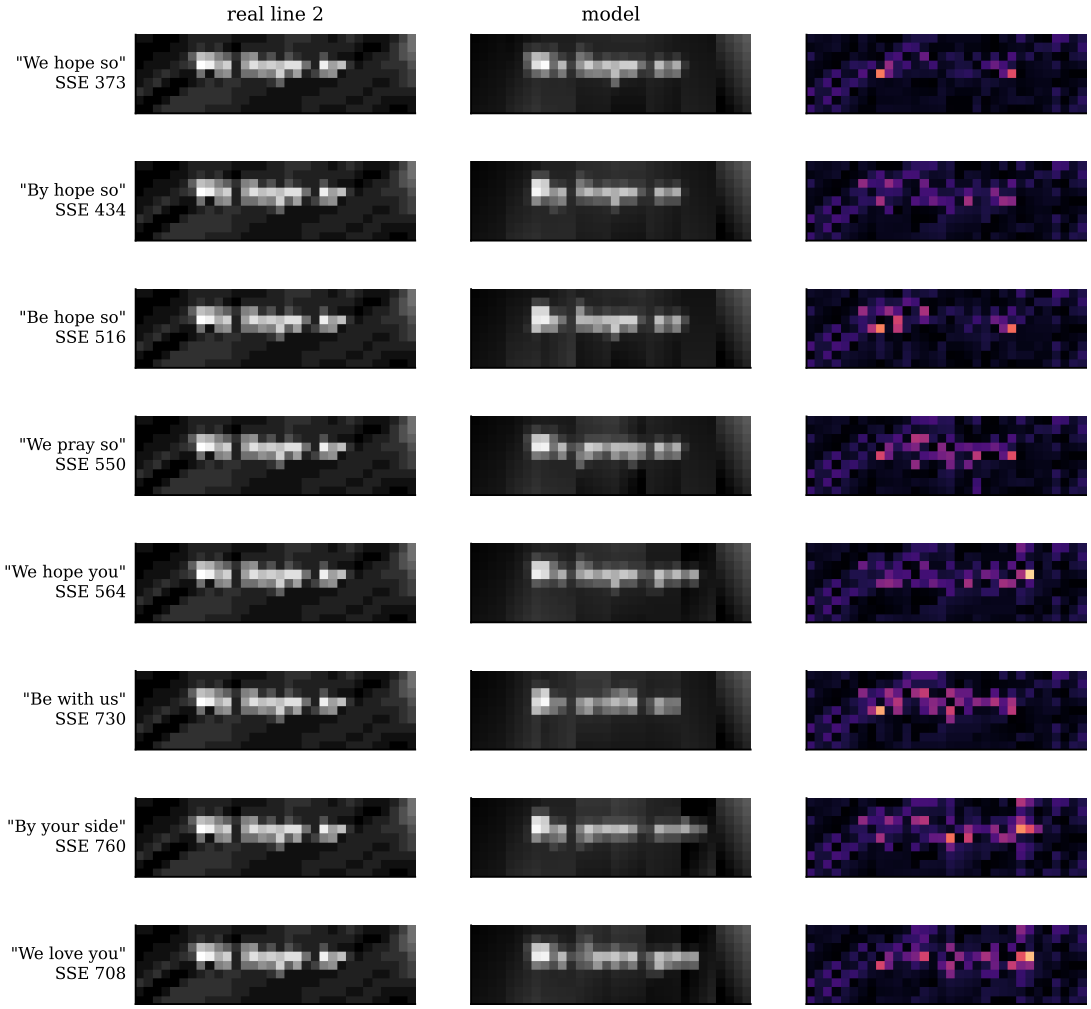



**Figure 19:** Line 2 for alternative sentences: model (middle) and absolute difference (right) next to the real plaque (left).

# 8 Working backwards from “hope”

For line 1 we assumed the text and solved for the process. We can do the same for line 2, using its most certain word.

**What may differ between the lines.** Canvas, resize filter, blur, gamma and ink colour are *shared*: both lines sit in one image, which is resized once. Each typed line may still have its own size, tracking, kerning, advance mode and even style. In Section 7 those were copied from line 1.

**Fitting line 2’s typesetting on “hope” alone.** We render only the word “hope” and score it only on its own pixels: the stone blocks covering texture columns 12–19. The search covers:

- four faces: Palatino Bold Italic and Italic, Apple and URW;
- sizes 13–21 px and tracking −1 to +3;
- kerning on/off and two advance modes;
- all positions.

Both Bold Italic faces fit equally well (44.5 and 45.2), at **18 px, tracking +0 (line 1: +1), whole-pixel advances and no kerning** (Fig. 20). The Italic (non-bold) faces fit worse.

**Then judging “We” and “so” on their own pixels.** This is a held-out test. The typesetting was calibrated on columns 12–19 only, and the first word and the ending are then scored on *different* pixels: columns 0–11 and 20–31 (Figs. 21 and 22).

- **First word (URW line 2).** “We” is the best real word (179). It is ahead of “Me” (202); “By” drops out of contention. Only the non-word “Wr” is lower (161).
- **First word (Apple line 2).** The W-forms lead again.
- **“We’ll”.** With the calibrated setting, “We’ll” matches its own region best of all, because its extra strokes resemble a faint mark near column 9. On the whole line, however, “We’ll hope so” scores 642 (URW), against 329 for “We hope so”. The wider word pushes “hope so” out of place. “We’d”, “Well”, “Will” and “We will” fail in the same way (544–1000). “By” shows the same effect, so a good local fit alone is not enough.
- **Ending. “so” becomes the best real ending in both faces**: 106 vs. “or” 118, “us” 120, “he” 124 (URW); 107 vs. “as”, “or” 115, “us” 118 (Apple). Only non-words such as “nr” or “se” are lower.
- **Whole line (URW).** “We hope so” becomes the best of all variants (338), ahead of “We hope or” (341), “We hope us” (343) and “We hope he” (346). It is also the best line-2 error of the whole study (Fig. 23).

**The unconstrained search now finds the sentence on its own.** We repeated the global dictionary search under the calibrated setting: 17.7k words, any number of words, any position, no hints (Fig. 24). The best strings are now “We hops So”, “Wt hope so” and **“We hope so”** (rank 3, 328.7; URW). With Apple Palatino the top is “We hops So” with “We hope So” at rank 3.

- Without any guidance, the search now produces *We* at the start and *so* at the end. Under line 1’s typesetting it had preferred “he”/“hr” endings.
- The only rival for the middle word is “hops”, which differs from “hope” by one small letter. That is at the limit of resolution; “hope” is the English choice.
- Upper- or lower-case “so” cannot be distinguished.

**Why this matters.** Calibrating the typesetting on one word made the other two words come out without being told. This is the same self-consistency that identified “Eternal Star”: one process and one font explain the whole line, and they predict “We” and “so” on pixels that were not used to fit them.

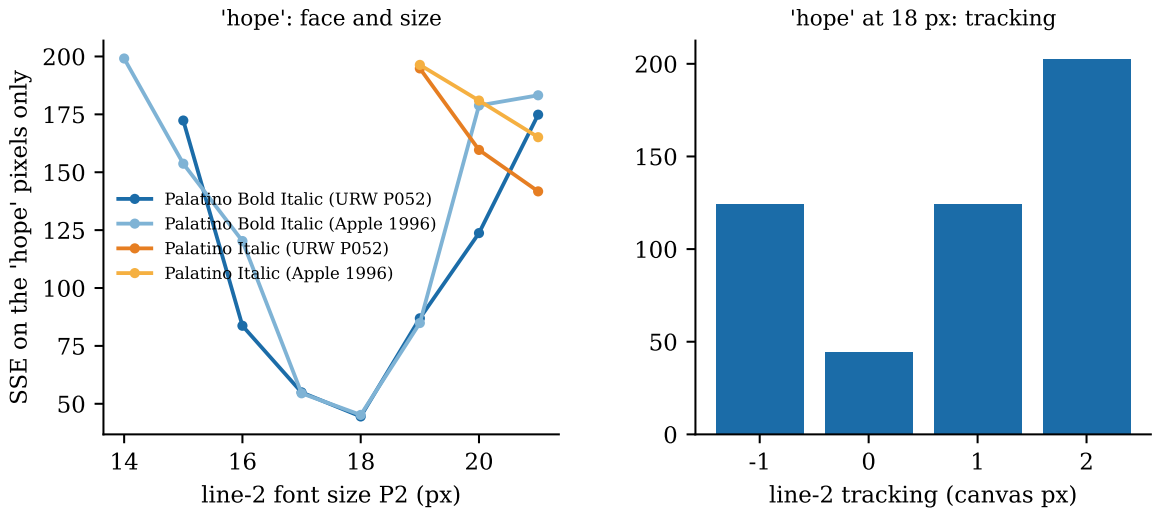



**Figure 20:** Line-2 typesetting fitted on the four letters of “hope” only. Left: error vs. font size for four faces. Right: error vs. tracking at the best size.

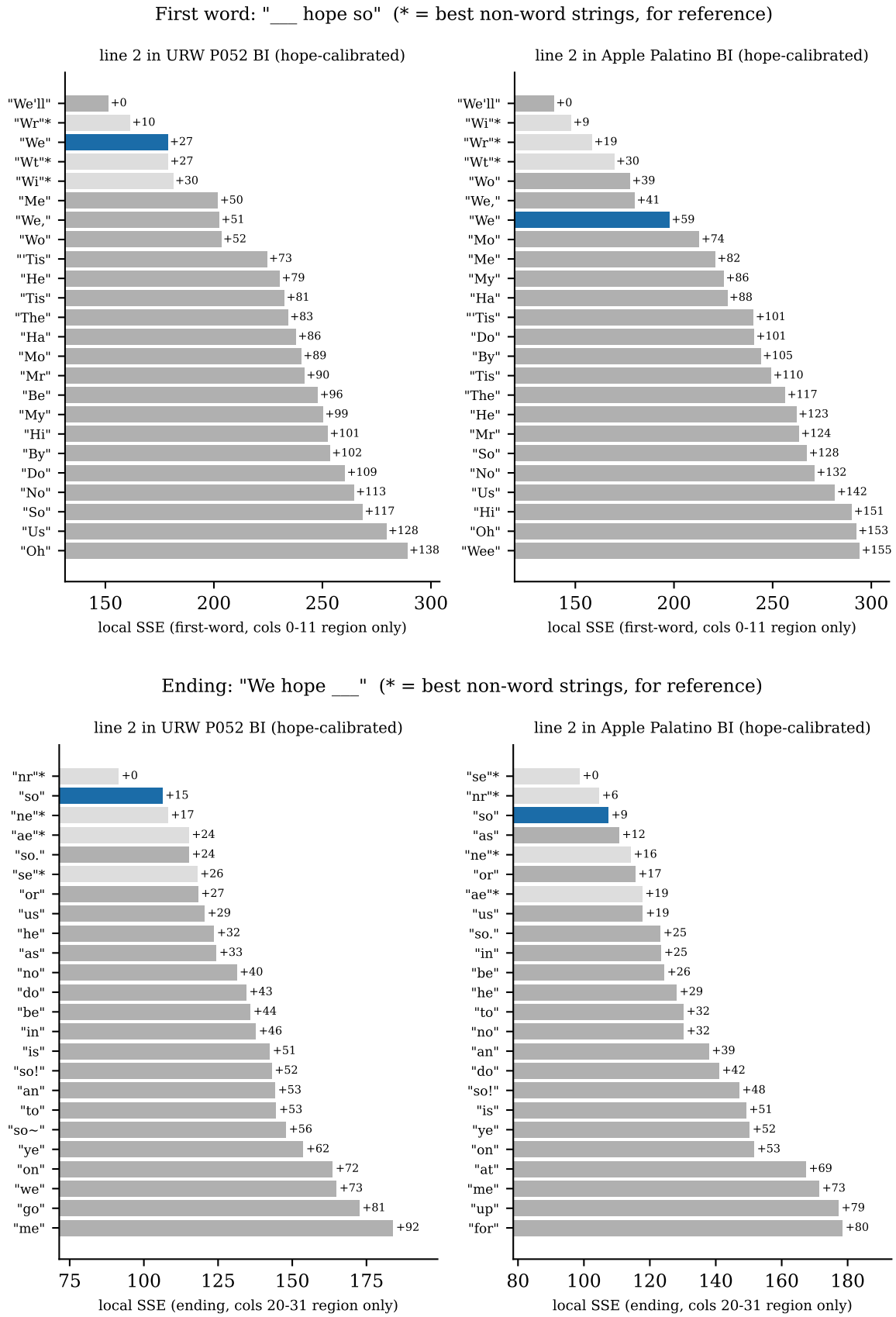



**Figure 21:** After calibrating line 2 on “hope”: local errors of first-word candidates (top) and ending candidates (bottom).

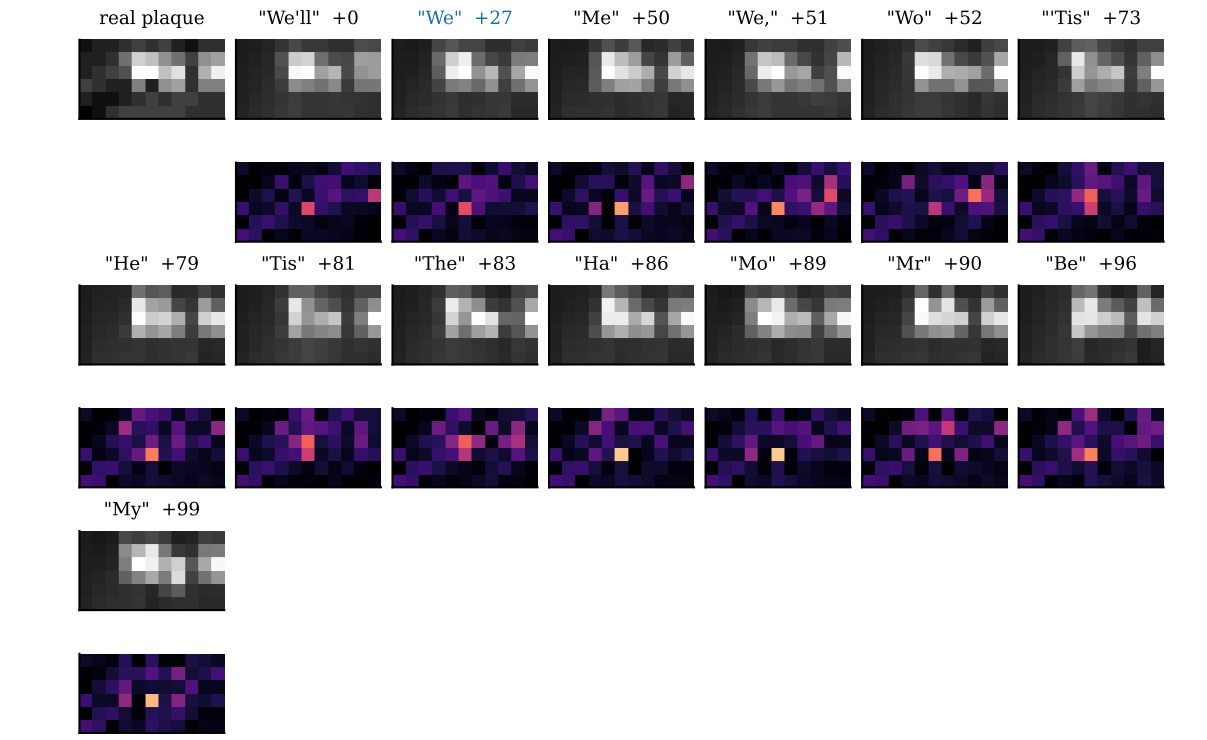



**Figure 22:** Close-ups after the hope calibration (URW line 2): first word (top) and ending (bottom). Model above, absolute difference below.

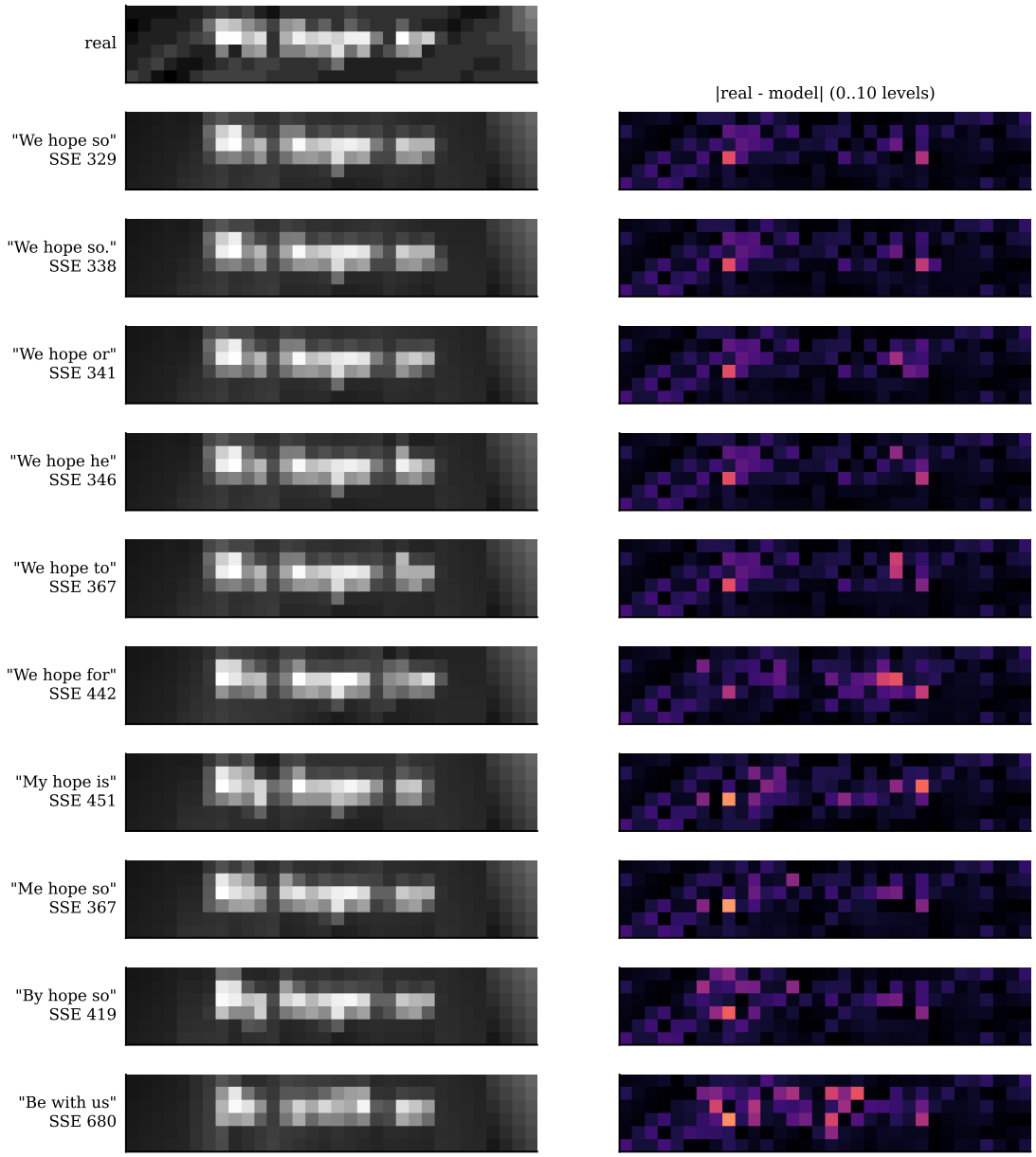



**Figure 23:** Whole-line variants under the hope-calibrated line-2 typesetting (URW): real plaque (top), model (left) and absolute difference (right).

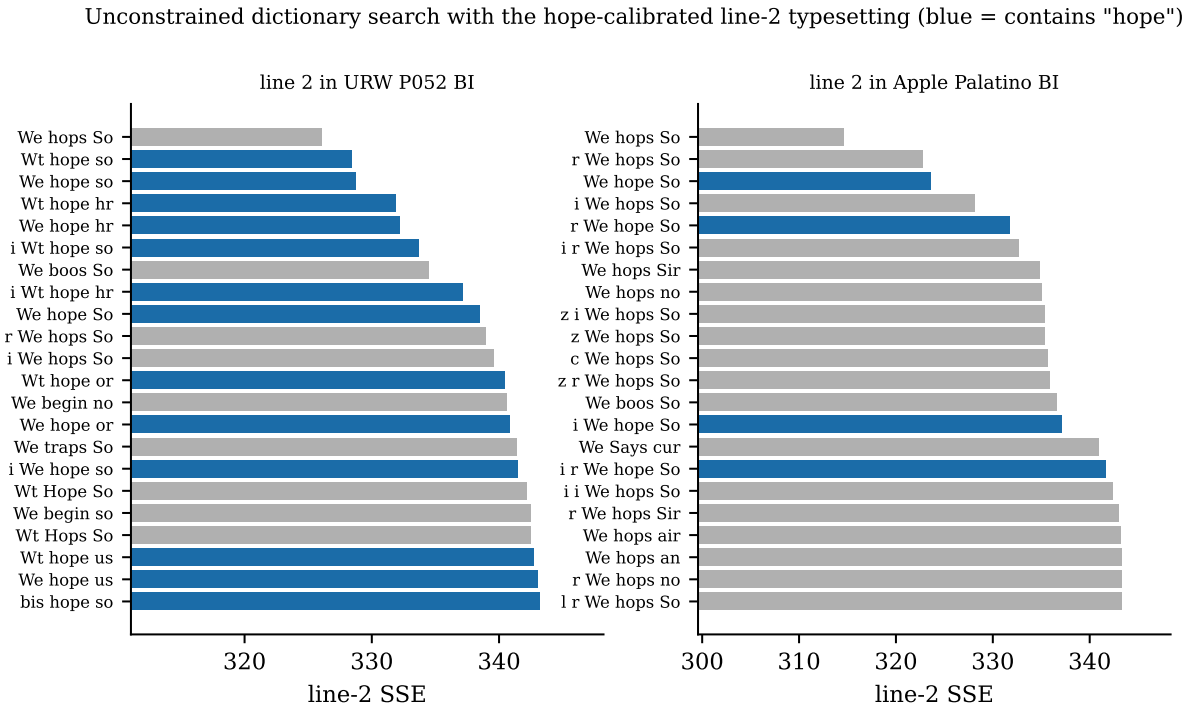



**Figure 24:** Unconstrained dictionary search under the hope-calibrated line-2 typesetting. "We . . . so/So" now emerges on its own.

# 9 Calibration: how much can the pixels tell?

We planted 24 known phrases into the real plaque. For each one we replaced the ink of the best reading by the phrase's ink and kept the real stone and the real fit residual. We then decoded each plaque blind with the same dictionary search (Fig. 26).

- **Exact recovery is rare.** The exact phrase came back first in 2/24 cases, and 32/68 words (47 %) were recovered. With a large vocabulary, some nonsense string often fits slightly *better* than the truth.
- **The truth is never far from the optimum.** The true phrase always scored within ≈ 58 SSE of the best string found. This defines a *pixel-compatible window*: readings more than about 58 above the optimum are effectively excluded, and readings inside it are distinguished by language.
- **Strong words survive.** "hope" (or "hops"), "Shine", "Thank you", "happy" and "light" were recovered well. Short function words and two-letter endings were not.
  Under line 1's typesetting, the optimum for the real plaque is about 341–344.
- **Inside the window, in both fonts:** "We hope so" (373 Apple, 396 URW).
- **Well outside:** "My hope is" (459–511), "No hope so" and "'Tis hope so" (466–474), and "lie here ~" (689–695).

**Calibration under the hope-calibrated typesetting.** We repeated the test with line 2's final typesetting from Section 8 (Fig. 25).

- 4/24 phrases were recovered exactly and 40 % of words were recovered.
- The truth always lay within 52 of the best string found.
- On the real plaque, "We hope so" lies only **2.6** above the global optimum (328.7 vs. 326.1 for the non-English "We hops So"). That is deep inside the window where, in simulation, the true text is found.

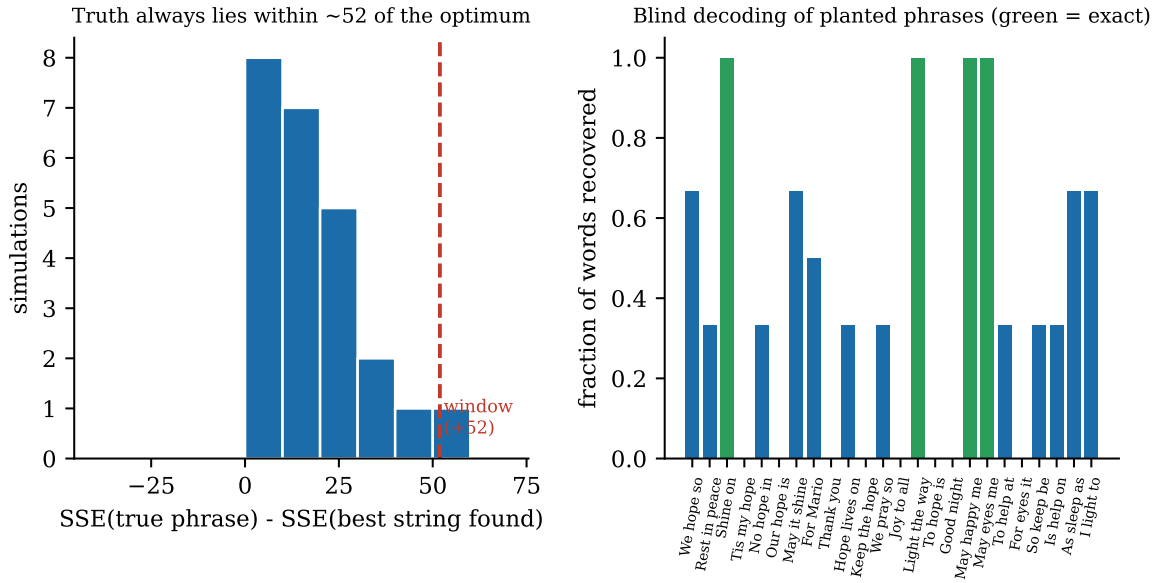



**Figure 25:** Calibration under the hope-calibrated line-2 typesetting.

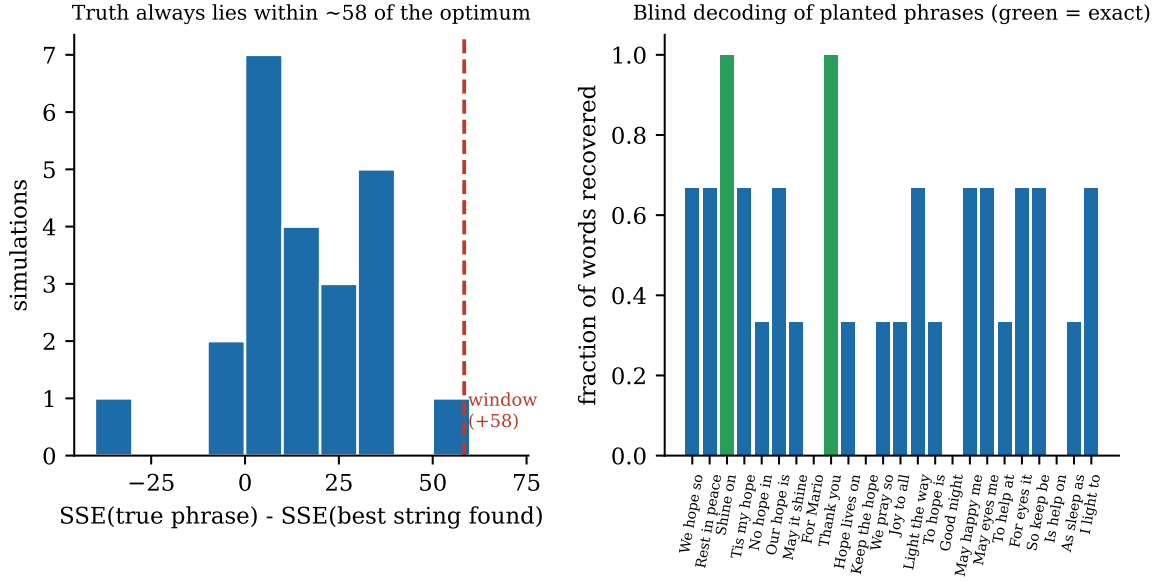



**Figure 26:** Calibration on synthetic plaques with known text: the gap between the true phrase and the best string found (left), and the words recovered per simulation (right).

# 10 The artist's font file: Adobe's 1990 PostScript Palatino

The family is settled. We then asked which *file* the artist used. Every digitisation of Palatino we could obtain went through the identical pipeline: line-1 recipe search, then line 2 calibrated on "hope" (Table 5 and Fig. 27).

- **URW Palladio**, the 1990s PostScript clone, in three generations: Ghostscript 5.10 (1998) and 6.0 (1999) from the GNU archive, and TeX Live's Palladio L.
- **TeX Gyre Pagella** and **URW P052**.
- **Apple's System 7.5.3 Palatino** (1996).
- **Palatino Linotype** (1998).
- **Adobe's own Type 1 Palatino** (`Palatino-BoldItalic 001.005`, dated 2 July 1990), from an *Adobe Type Basics* disc of 1994. This copy was supplied by the reader and is not redistributed.

**Adobe fits best.** Adobe's original fits best on both lines: line 1 731.3, line 2 311.2, total 1042.5 against 1085.1 for the best clone, an improvement of 43.

- **Recipe.** 192×160 canvas, 26 px, tracking +2, blur 0.30, gamma 1.4, at 4.41 times the noise floor, the closest any rigid model came.
- **Only Palatino benefits.** The other genuine Adobe fonts on the same disc (Times, New Century Schoolbook, Garamond, Zapf Chancery, Bookman) do *not* beat their clones. The Adobe advantage is specific to Palatino.
- **Unhinted rendering is consistently worse.** The original was rasterised *with* hinting, as Adobe Type Manager did in 1996.

**The blind search now finds the sentence by itself.** With Adobe's Palatino, the unconstrained dictionary search over 17.7k English words returns exactly **"We hope so"** as its single best string, in both settings (Fig. 28):

- line 1's typesetting: 362.1, ahead of "We hope he";
- hope-calibrated line 2: 311.2, ahead of "We hope as".

With every other font the top result had been a non-word such as "Wt hope he". On its own pixels, "so" is now the best of *all* endings, non-words included (91.4 vs. "nr" 92.7). We conclude that the artist most likely typeset the plaque with Adobe's PostScript Palatino Bold Italic, as found on Macintoshes with Adobe Type Manager in the mid-1990s.

**One pixel the fonts cannot explain.** At the bottom of the "W" the real plaque shows bare stone (row 19, column 8; value 8) between two bright points (Fig. 29). Every font predicts ink there, Adobe included (14.8). We tested and rejected global explanations:

- full 1990s TrueType hinting;
- sharpening after the resize;
- a drop shadow or emboss;
- 1996 Mac per-size advance widths;
- unhinted rendering.

Moving only the W by one artwork pixel improves line 2 markedly (311 → 281 with Adobe) but does not empty the hole. A second, similar hole sits between "Ete" and "rnal" on line 1. The most plausible explanation is the artist's hand: manual kerning or a one-pixel touch-up of the finished 32×32 texture, both common practice for tiny text.

| digitisation | line 1 | line 2 | total | hole (19,8) |
|---|---|---|---|---|
| Palatino Bold Italic (Adobe Type 1, 1990) | 731.3 | 311.2 | 1042.5 | 14.8 |
| URW Palladio BI (Ghostscript 6.0, 1999) | 754.9 | 330.2 | 1085.1 | 14.1 |
| URW Palladio L BI (TeX Live) | 754.9 | 330.2 | 1085.1 | 14.1 |
| Palatino Bold Italic (URW P052) | 756.9 | 329.0 | 1085.9 | 14.2 |
| URW Palladio BI (Ghostscript 5.10, 1998) | 755.1 | 332.2 | 1087.4 | 14.1 |
| TeX Gyre Pagella BI | 756.6 | 331.5 | 1088.1 | 14.0 |
| Palatino Bold Italic (Apple 1996) | 759.2 | 350.4 | 1109.5 | 13.8 |
| Palatino Italic (Apple 1996) | 809.1 | 329.3 | 1138.4 | 14.3 |
| Palatino Linotype BI (1998) | 793.8 | 367.3 | 1161.1 | 15.2 |
| URW Palladio Italic (gs 5.10) | 804.3 | 391.6 | 1195.9 | 14.1 |
| Palatino Italic (Adobe Type 1) | 802.9 | 395.7 | 1198.5 | 14.1 |

**Table 5:** Palatino digitisations under the identical pipeline (hinted rendering). Lower is better. The last column is the predicted value at the W hole; the real plaque shows 8.

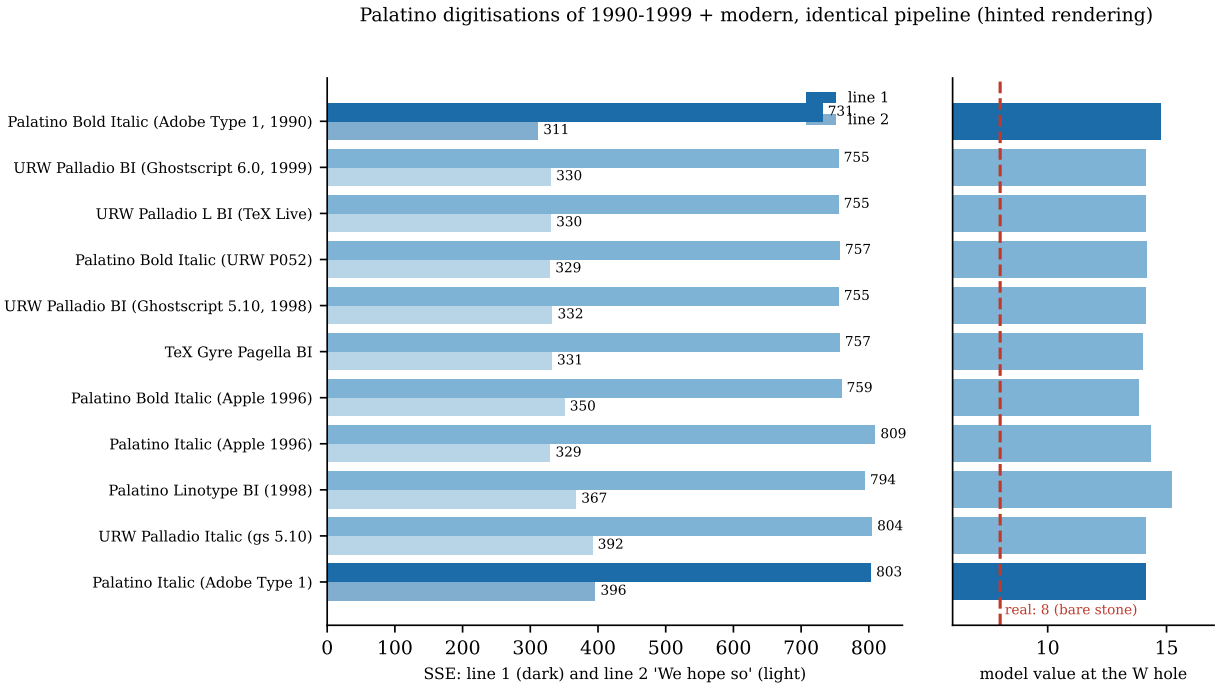



**Figure 27:** Line-1 and line-2 errors for each Palatino digitisation (left) and the predicted value at the W hole (right).

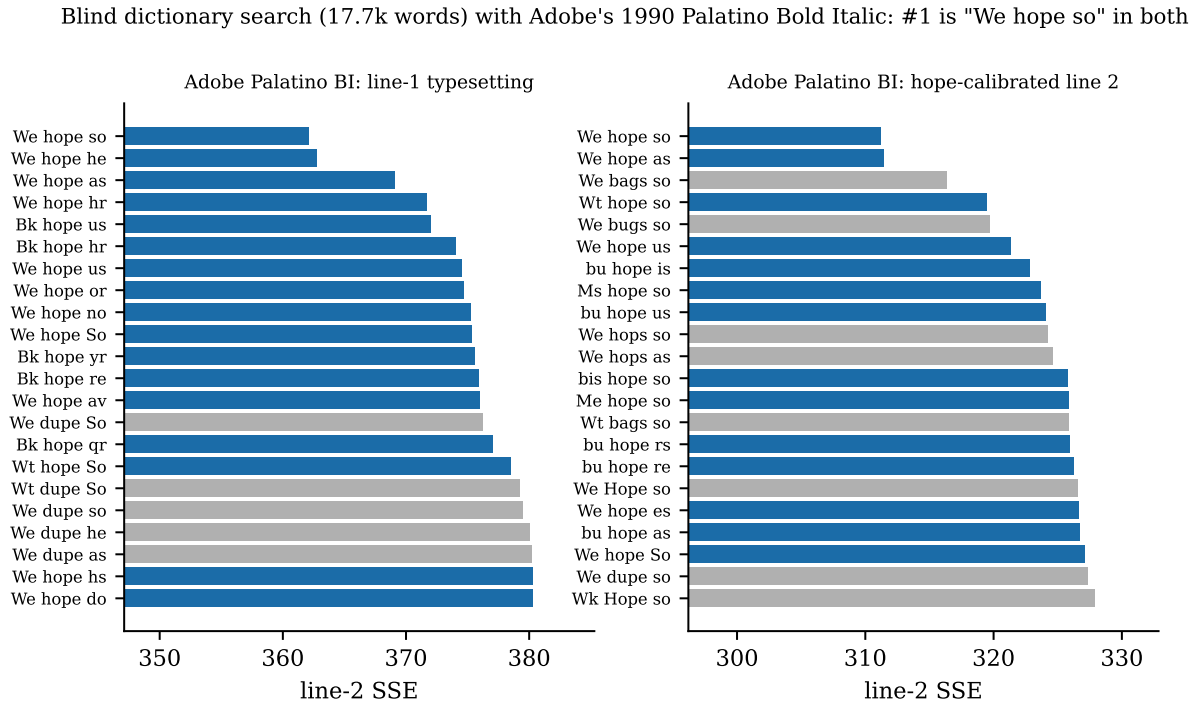



**Figure 28:** Blind dictionary search with Adobe's 1990 Palatino Bold Italic. "We hope so" is the single best string of the whole search, with line 1's typesetting (left) and with the hope-calibrated line 2 (right).

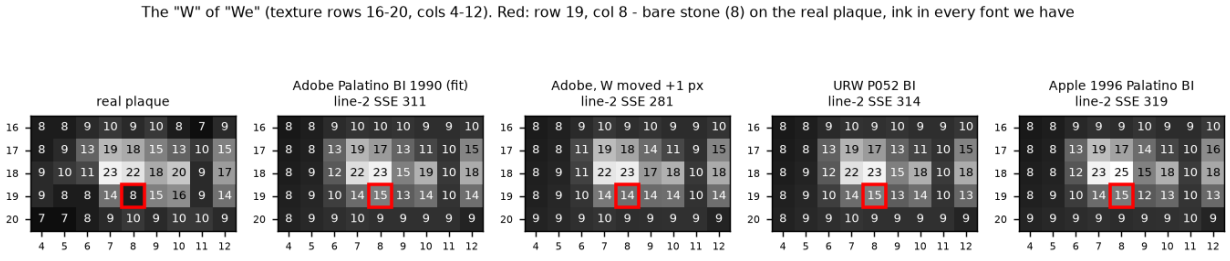



**Figure 29:** The W of "We", texture rows 16–20, columns 4–12, with grey levels printed. The red box marks the pixel that is bare stone on the real plaque but inked in every font.

# 11 Inverting the process: can the text be made visible?

The forward model maps a high-resolution artwork to the 32×32 texture, so it can also be run backwards. We solve for the 160×128 artwork whose rendering, after the fitted resize, blur, gamma, ink level and stone removal, reproduces the plaque.

**Why priors are needed.**

- **Without a prior the problem is wildly underdetermined.** The inverse reproduces the texture *exactly* (residual 0), including its stone noise, and looks like noise (Fig. 30, column 2).
- **Prior used.** We therefore add a sparsity and smoothness prior: most of the artwork is empty, and ink comes in strokes. Optionally the ink is also pushed to be black or white.
- **Setting its strength.** By the *discrepancy principle*: the leftover error must equal the measured noise floor of the two bands (331). The solver may then explain the ink, but not the noise.

**Control plaques.** A prior can draw letter-like strokes on its own, so we apply the *identical* inversion to synthetic plaques with known text.

- **Construction.** Every hypothesis gets the same smooth stone and the same real stone noise, taken from the clean rows 23–31. For "L is real 2401", row 2 keeps the real pixels because that claim says nothing about it.
- **Text is recoverable in principle.** On these plaques the inversion returns legible text ("Eternal Star", "L is real 2401", "A Secret star"). A correctly modelled plaque at this noise level could therefore be partly "read" after inversion.
- **The real plaque is less legible.** It inverts to letter-like blobs with the word structure "E ternal star / We hope so". The cause is not noise but the remaining model mismatch of Section 5: the real 1996 letter spacing and rasterising differ slightly from our fonts.

**Similarity to the hypotheses.** The correlation between the real inversion and each synthetic inversion, in the artwork space, is:

- row 1: **0.74** for "Eternal Star" vs. 0.29 for "L is real 2401" and 0.30 for "A Secret star";
- row 2: **0.76** for "We hope so" vs. 0.68 for "My hope is" and 0.44 for "lie here ~".

The inverted plaque looks like the inversion of "Eternal Star / We hope so" and clearly unlike the published alternatives. This is consistent with, and independent of, the forward scoring in the previous sections.

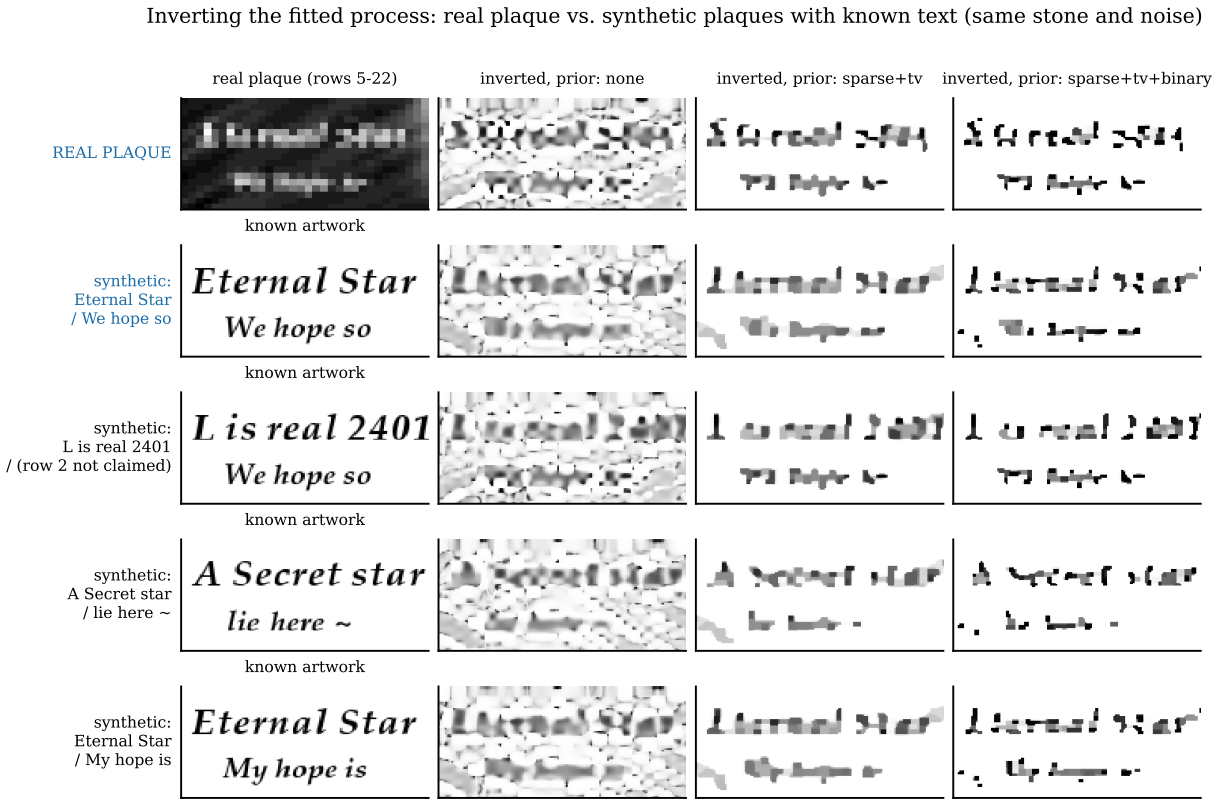



**Figure 30:** Inverting the fitted process. Top row: the real plaque and its inversion with three priors. Other rows: synthetic plaques with known artwork (left) on the same stone and noise, inverted identically. Column 2 (no prior) reproduces the pixels exactly but is meaningless. Columns 3–4 use the prior strength that leaves exactly the noise floor.

## 12 Comparison with published readings

Every claim was scored with the same model. Each claimed line 1 received its *own* exhaustive recipe search. Line 2 was scored jointly under that recipe, and also under the "Eternal Star" recipe. The results are in Table 6, Fig. 31 and Fig. 32.

**"L is real 2401".** The legend puts the whole phrase on row 1 and makes no claim about row 2, so we scored it that way. Row 1 received its own exhaustive recipe search, and row 2 was not scored.

- **Row 1 fit.** Even so, "L is real 2401" fits row 1 about twice as badly as "Eternal Star": 1492 (Apple) against 759. The "2041" spelling does the same (1561).
- **Why.** The texture shows a long, evenly inked first word with a tall capital at the start and no gap after it. "L is" predicts a short capital L and a two-letter word separated by spaces, and the digits predict wide, cap-height shapes where the texture has italic lower-case letters (Fig. 32).
- **Row 2.** Its text is left unexplained by this reading.

Our results agree with the 1998 Nintendo letter that the legend has no basis in the texture [4].

**"A Secret star / lie here ~" (Nodd).** Nodd correctly identified "star" and a three-part line 2. However, "A Secret" fits line 1 far worse than "Eternal", and "lie here ~" has no descender where the texture has one.

**"Eternal Star" (Super Mario Wiki).** Confirmed. It is the best line 1 of all, and its "heavily serifed font" [1] is Palatino Italic.

**"My hope is".** This reading gets "hope" right, but it predicts a second descender, the "y", which is absent (Fig. 13). It is also clearly worse than "We hope __" in both fonts (Table 6).

| line 1 | line 2 | proposed by | line-1 SSE | | line-2 SSE | |
|---|---|---|---|---|---|---|
| | | | Apple | URW | Apple | URW |
| “L is real 2401” | (not claimed) | 1990s rumour (‘Luigi is real’), whole phrase on row 1 | 1492 | 1414 | – | – |
| “L is real 2041” | (not claimed) | variant spelling, row 1 (e.g. MobyGames trivia) | 1561 | 1437 | – | – |
| “A Secret star” | “lie here ~” | Daniel Nodd (No64DD), Nintendo Life | 1604 | 1460 | 1126 | 1320 |
| “L is real star” | (not claimed) | reader comment, Nintendo Life | 1421 | 1378 | – | – |
| “Eternal Star” | (not claimed) | Super Mario Wiki (‘more likely’) | 759 | 757 | – | – |
| “Eternal Star” | “My hope is” | proposed reading (two descenders + ascender) | 759 | 757 | 511 | 459 |
| “Eternal Star” | “my hope is” | same, lower case | 759 | 757 | 489 | 475 |
| “Eternal Star” | “lie here ~” | Nodd’s line 2 under Eternal Star | 759 | 757 | 695 | 689 |
| “Eternal Star” | “We hope so” | this work | 759 | 757 | 373 | 396 |
| “Eternal star” | “We hope so” | this work (lower-case s) | 765 | 755 | 554 | 565 |

**Table 6:** Published and proposed readings. Lower is better. Line-1 noise floor 165.7. The line-2 pixel-compatible window is roughly 344–402.

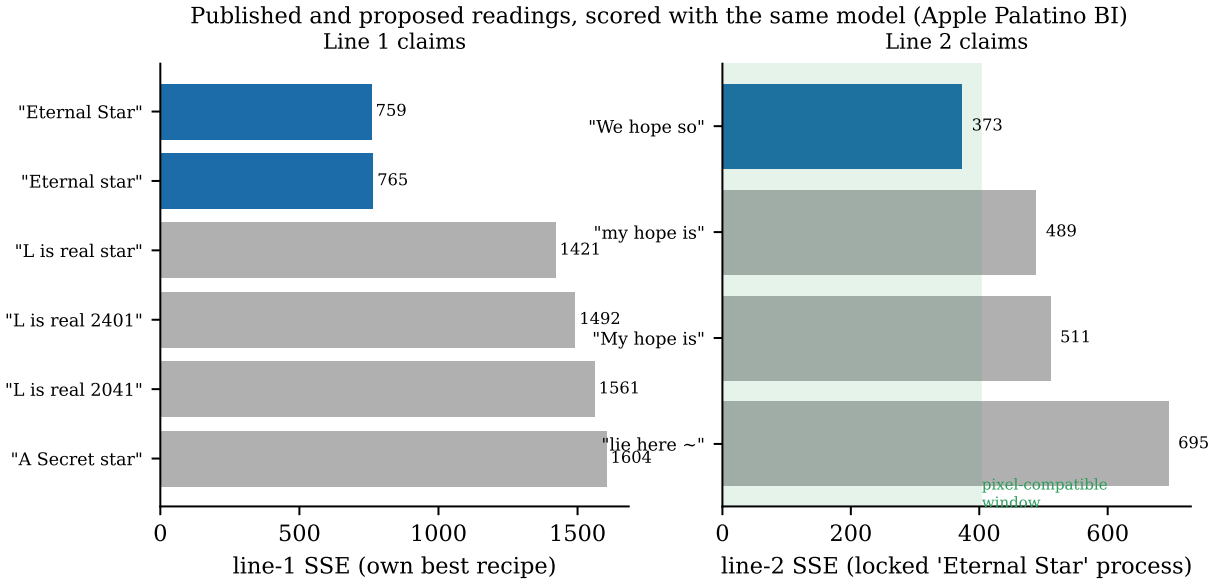



**Figure 31:** Line-1 and line-2 errors of the claims. The green band is the line-2 pixel-compatible window.

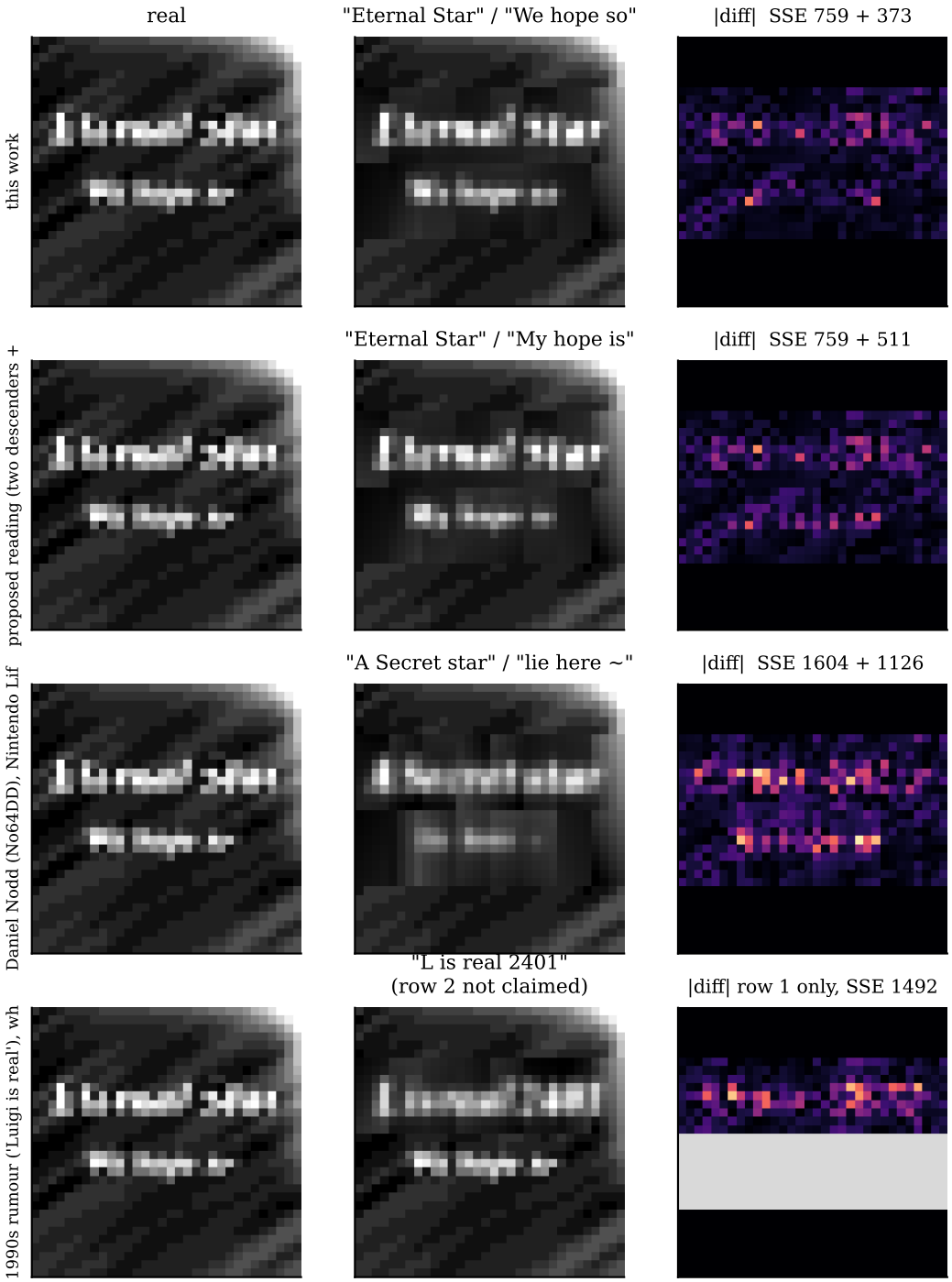



**Figure 32:** Whole-plaque reconstructions of four readings: real plaque, model rendering and absolute difference.

# 13 Limitations

- **Rasteriser.** The exact 1996 rasteriser (hinting, letter spacing) is unknown. This is the source of the remaining gap to flexible fits, about 4.6 times the noise floor.
- **Stone model.** The stone under the letters is modelled, not known.
- **Two-letter ending.** Under line 1's typesetting the ending cannot be resolved by the pixels. After the hope calibration "so" is the best real ending, but its margin over "or", "us" and "he" is small, and meaning still supports the choice.
- **Case.** Capital vs. lower-case "S" in "Star" is undecided: line 1 alone cannot separate them, and the joint two-line test points in opposite directions for the two fonts. Whether the final word is "so" or "So" cannot be decided either.
- **Vocabulary.** The dictionary searches are only as complete as their vocabulary. Very rare words, names and non-English text were not covered exhaustively.
- **Separate line fits.** The process is fitted on line 1 and then shared with line 2. In one joint test (URW), a slightly different process fitted line 2 much better (283) at a small cost on line 1. A fully joint two-line recipe search could tighten the model further. It would not change the reading: it was run with "We hope so".
- **No source material.** We did not use original source art or leaked material. Such material, or a statement by the original artist, would be the only way to reach certainty.

# 14 Conclusion

A physically motivated forward model, searched exhaustively and calibrated on synthetic plaques, shows that the plaque was made by typing two centred lines in **Palatino Bold Italic**, most likely Adobe's 1990 Type 1 version, on a 160×128 canvas, at 23 px (line 1) and 17–18 px (line 2). The artwork was resized to 32×32 with a bilinear filter, slightly blurred and gamma-adjusted (1.8).

Our confidence in each word:

| word | evidence | confidence |
|---|---|---|
| Eternal | every font, process and test | very high |
| Star / star | rank 1 of 27,122; rivals with their own recipes lose; case undecided (the two fonts disagree) | high (word), open (case) |
| hope | rank 1 of 12,556; appears unprompted in every global search; its letters calibrate line 2 | high |
| We | best English word of the required shape; best real word on held-out pixels after the hope calibration; "By", "We'll" and "Me" fail on the whole line | high |
| so | two small letters are certain; best real ending after the hope calibration and the only one that completes the sentence; margin to "or", "us" and "he" is small | moderate |

Running the fitted process backwards gives an independent visual check. The inverted plaque resembles the inversion of "Eternal Star / We hope so" far more than that of "L is real 2401" or "A Secret star / lie here ~" (Section 11).

The most likely text of the plaque is ***"Eternal Star — We hope so"***. That is a light-hearted line, in keeping with the 1998 description of the plaque as a developers' joke, and certainly not "L is real 2401".

# A Reproducing the results

The code is released as the Python package `plaquedecode` (MIT licence). No ROM, texture or commercial font is distributed.

```
pip install -r requirements.txt
python -m plaquedecode extract --rom "Super Mario 64 (USA).z64"     # -> texture.npy (local)
# put P052-BoldItalic.otf (free, URW base35) into fonts/
R=results/recipes_P052-BoldItalic.json
python -m plaquedecode recipe   --font P052-BoldItalic.otf --stage full    # line-1 process
python -m plaquedecode word2    --recipes $R                               # word 2 of line 1
python -m plaquedecode line2    --recipes $R --mode dp                     # dictionary search
python -m plaquedecode line2    --recipes $R --mode combo --phrases curated
```

```
python -m plaquedecode candidates --recipes $R --line2 results/line2_dp_P052-BoldItalic.json
python -m plaquedecode hopefit  --recipes $R --line2 results/line2_dp_P052-BoldItalic.json \
                                --fonts2 P052-BoldItalic.otf
python -m plaquedecode simulate --recipes $R --line2 results/line2_dp_P052-BoldItalic.json
python -m plaquedecode claims   --font P052-BoldItalic.otf --recipes $R
python -m plaquedecode invert   --recipes $R --hopefit results/hopefit_PalatinoBoldItalic.json \
                                --line2 results/line2_dp_hopefit_P052line2.json
python -m plaquedecode figures  --out report/figures                    # + report/tables/
```

The searches take minutes to an hour on a consumer GPU. All result files used for this report are included in `results/`, so the figures and tables can be regenerated without a GPU.

# References


[1] Super Mario Wiki, *Courtyard (Peach's Castle)*. https://www.mariowiki.com/Courtyard_(Peach%27s_Castle)

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[3] MobyGames, *Super Mario 64 – trivia*. https://www.mobygames.com/game/wii/super-mario-64/trivia

[4] My Nintendo News, *Nintendo finally explains the Super Mario 64 fountain mystery*, 11 Dec 2016. https://mynintendonews.com/2016/12/11/nintendo-finally-explains-the-super-mario-64-fountain-mystery/

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[7] n64decomp, *Super Mario 64 decompilation*. https://github.com/n64decomp/sm64

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